

A Project Report on
Energy-Efficient True Single Phase Clocked Flip Flop
for High Performance Applications

Submitted in the partial fulfillment of the requirements for the award of the degree of

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APRIL 2025



CERTIFICATE

This is to certify that the project entitled “**ENERGY-EFFICIENT TRUE SINGLE PHASE CLOCKED FLIP FLOP FOR HIGH PERFORMANCE APPLICATIONS**” that is being submitted by

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DECLARATION

We hereby declare that the project work entitled “**AUTOMATIC USER STATE RECOGNITION FOR HAND GESTURE BASED LOW COST TELEVISION CONTROL SYSTEMS**” is being submitted to **Vignan’s Institute of Engineering for Women (A)**, Affiliated to **JNTU-GV** University, in partial fulfillment for the award of B.Tech degree during the year 2021-2025 in **Electronics and Communication Engineering**. The work was originally designed and executed by us under the guidance of our supervisor **Mrs. B. Manjula** and was not a duplication of work done by someone else. We hold the responsibility of the originality of the work incorporated into this thesis.

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ABSTRACT

Flip-flops (FFs) are fundamental components in digital circuits, playing a crucial role in data storage and synchronization. Optimizing their control mechanisms can significantly reduce power consumption in advanced systems. A specialized True-Single-Phase-Clocked (TSPC) flip-flop has been designed to operate with lower energy consumption. This optimized TSPC flip-flop incorporates an input-aware pre-charge strategy, which ensures that pre-charging occurs only when necessary, thereby minimizing unnecessary power dissipation. Additionally, transistor-level optimizations and floating node analysis have been employed to further enhance energy efficiency without significantly increasing the circuit's area. These optimizations help reduce leakage power and unnecessary switching activity, leading to improved performance. At a supply voltage of 1V, the proposed flip-flop achieves a 17.6% reduction in CK-to-Q delay compared to a Transmission Gate Flip-Flop (TGFF). This reduction in propagation delay improves the overall speed and efficiency of digital systems, making the proposed design particularly beneficial for low-power VLSI circuits, battery-powered devices, and high-speed processors where energy efficiency is critical.

Keywords:- True-Single-Phase-Clock(TSPC),D-Flip-Flop, Transmission-Gate Flip - Flop, (TG FF),Single Event Upset (SEU) and Single-Event-Transient (SET).

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CHAPTER 1

INTRODUCTION

1.1 FLIP FLOPS

Most high-performance digital designs today utilize a synchronous clock to order events. The principle of synchronization is easy from the perspective of system design. However, ordering all events in a high-performance design in a synchronous fashion requires generation and distribution of clock signals at multi GHz clock frequencies, which is challenging. Therefore, it is very important to design the flip-flops so that they can be optimized for the yield and reliability constraints. Synchronization circuits such as latches and flip-flops constitute the clocked registers that synchronize the data flow in a VLSI circuit.

Hence, flip-flops and latches are among the most important circuit blocks in a digital synchronous chip design. Ideally, timing circuits like flip-flops and latches should add as little latency as possible, and have low power dissipation. In practice, however, clocked registers can actually consume a substantial fraction of the clock-cycle period, and dissipate a considerable portion of the total power.

Flip-flops are an important building block in modern digital VLSI systems. Some of the major usage areas of flip-flops are in registers, pipelines and state machines, ensuring sequencing of data. A flip-flop reads an input value, saves it for some time and then writes the stored value at the output. This procedure is completed, regardless of the subsequent change in the elements input value. Based on the comparison of the power breakdown for different elements in VLSI chips, latches and flip-flops are the major sources of the power consumption in synchronous systems.

Flip-flops have a direct impact on power consumption and speed of VLSI systems. When estimating the power dissipation of a system, flip-flops may be a major power consumption component. The basic types of flip flops include, D Flip-Flop, T Flip-Flop, JK Flip-Flop, and Set-or-Reset (SR) Flip-Flop.

Flip-flops are crucial in digital circuits as they are capable of storing information, functioning as memory. They switch their output depending on input signals and clock pulses. This makes them useful in time-based applications such as registers and counters, where data has to be stored, updated, or toggled at a particular moment.

1.2 VARIOUS TYPES OF FLIP FLOPS

The memory elements in a sequential circuit are called flip-flops. Binary information can enter a flip-flop in a variety of ways and gives rise to different types of flip-flops. The most common types of flip-flops are:

1. SR (Set-Reset) Flip Flop
2. D (Data or Delay) Flip Flop
3. T (Toggle) Flip Flop
4. JK (Jack Kilby) Flip Flop

The flip-flops are synchronous sequential circuits. Hence the circuits change their states only when clock pulses are present.

1.2.1. Design of SR Flip Flop

The SR flip-flop consists of a simple NOR flip-flop, two AND gates. The two outputs and gates remain 0 as long as the clock pulse (or CP) is 0, regardless of the S and R input value and the R input value. When the clock pulse reaches 1, information is executed from the S and R inputs to the base flip-flop. Both are set at $s = 1$ and $r = 1$, so the clock pulse event can move two releases in a short time. It is one of the conditions, but you enter it at the end of the set with a zero, or the input of the flip-flop depends on whether the time is greater than 1. The SR Flip-Flop circuit diagram is shown in Figure 1.2 and characteristics is tabulated in Table 1.1

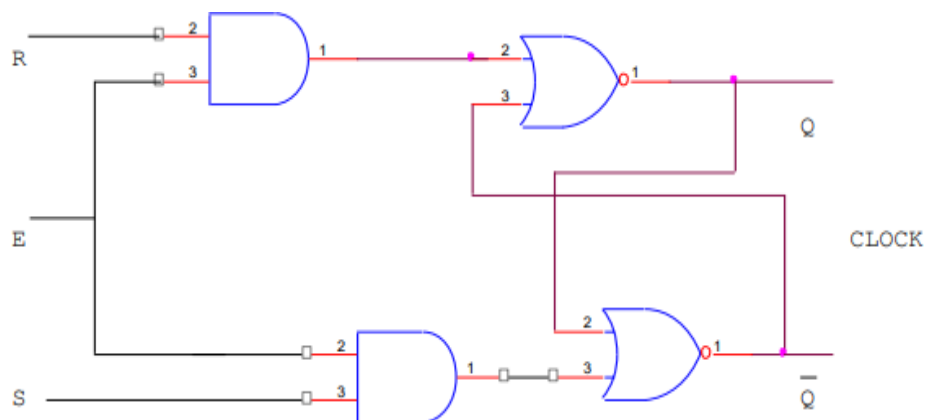


Fig 1.1: SR flip flop

Q	S	R	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	X
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	X

Table 1.1: Characteristic Table of SR flip flop

1.2.2. Design of D Flip Flop

RS Flip -One way to remove undesired states of indefinite conditions is to ensure that inputs s and R are not 1 at the same time. This is done with a D-flip-flop, as shown in Figure 1.3. The D flip-flop clock is a change of the SR flip-flop. D and CP: These are just two inputs. The D input goes directly into the entrance, and the D input is directed towards the complement-R input. The D input is a sample during the clock pulse. If 1, the flip-flop will switch to the specified state. If 0, the flip-flop is performed in a clear state. D-FF is one of the frequently used flip-flops. It can be static or dynamic.

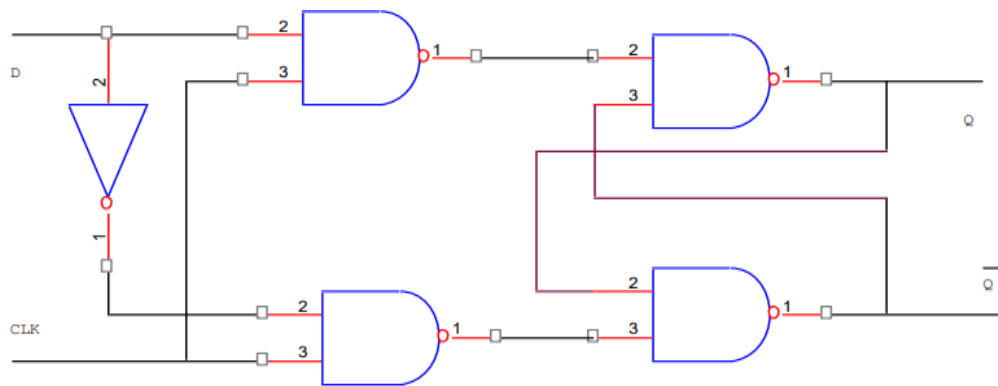


Fig 1.2: D flip flop

Q	D	Q(t+1)
0	0	0
0	1	1
1	0	0
1	1	1

Table 1.2.Characteristic Table of D flip flop

The data is input regarding the latest status of and flip-flop. Q and Q(t+1) represent the flip-flop output state. Based on the table, the output of the input is based on the state conversion. However, most importantly, this can only occur in the presence of a clock signal. This is exactly the same as the SR flip-flop in the free entry.

1.2.3. Design of JK Flip Flop

A JK flip-flop may be a refinement of the SR flip-flop in that the vague state of the SR sort is characterized within the JK sort. Inputs J and K carry on like inputs S and R to set and clear the flip-flop (note that in a JK flip flop, the letter J is for set and the letter K is for clear). When rationale 1 inputs are connected to both J and K at the same time, the flip-flop switches to its complement state, i.e., in the event that $Q=1$, it switches to $Q=0$ and bad habit versa. The Yield Q is ended with K and CP inputs so that the flip-flop is cleared amid a clock beat as it were in the event that Q was already 1. So also, yield Q' is ended with J and CP inputs so that the flip-flop is set with a clock beat it were in case Q' was already 1.

JK flip-flops are widely used in digital counters, frequency dividers, and shift registers due to their toggling feature.

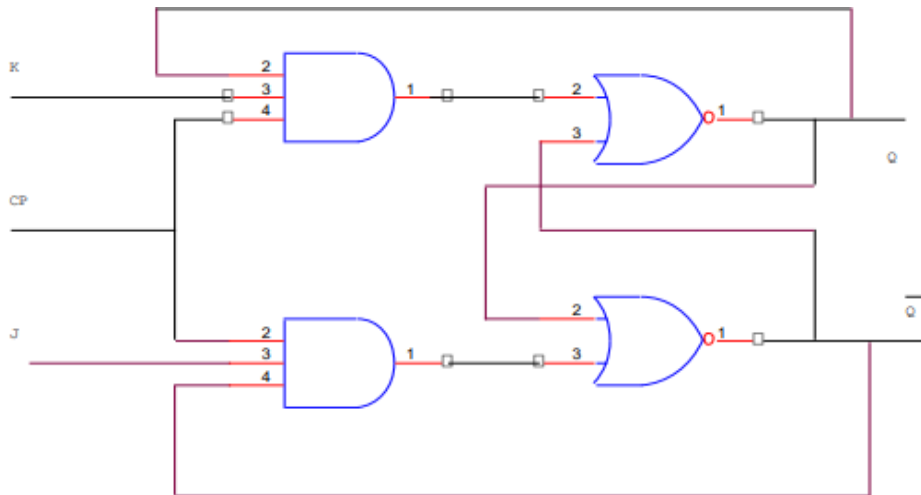


Fig 1.3: JK flip flop

Q	J	K	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

Table 1.3. Characteristic table of JK flip flop

It is exceptionally critical to realize that since of criticism association within the JK flip-flop, a CP flag which remains 1 whereas both J and K are break even with to 1 will cause the yield to complement once more and rehash complementing until the beat goes back to zero. To dodge this, the clock beats ought to be less than the proliferation delay through the time period of flip flops. Limitations on Beat width can be wrapped up with a master-slave or edge activated development.

1.2.3.Design of T Flip Flop

T flip flops could be a single input version of JK flip flops. T gets the flip tumble JK, at that point both the inputs are tied together. The assignment T comes from the capacity of the flip flounder to flip or complement, its state. In any case of the display state, the flip flounder complements its yield when clock beat happens whereas input $T = 1$.

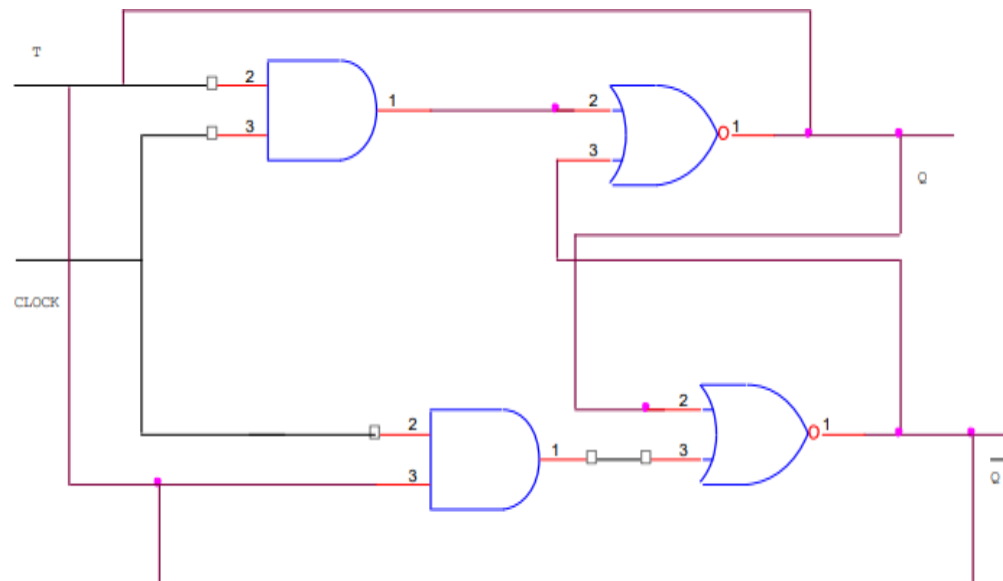


Fig 1.4: T flip flop

Q	T	Q(t+1)
0	0	0
0	1	1
1	0	1
1	1	0

Table 1.4. Characteristic table of T flip flop

1.3.1. Tri-State Driver

The tri-state driver is displayed by the symbol shown. The connection on the left is input and has an output on the character. The following connections are controls (or activated) that determine whether the output is powered on with an input value. The circles associated with this indicate that this control is low, that is, that H. is 0 V and the TRI-STATE driver is active for this input.

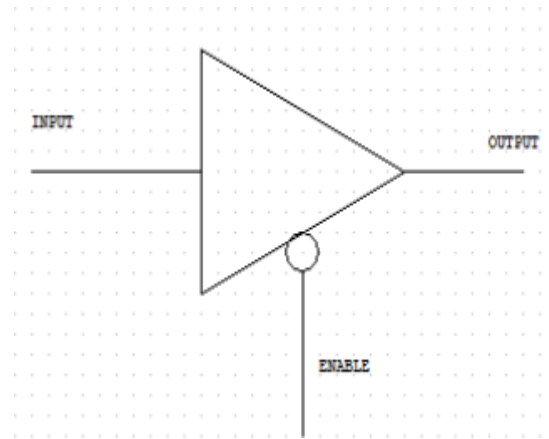


Fig 1.5: Tri-state driver

Control	Input	Output	Comment
0	0	0	Switch closed
0	1	1	Switch closed
1	0	x	Voltage on output not
1	1	x	Controlled by device

Table 1.5. Tri-state driver

The primary 2 columns appear the yield taking after the input when the driver is empowered. The final two lines demonstrate that when the driver is impaired, the abdicate is not under the driver's control; any additional circuitry connected to this surrender must be evaluated in order to determine its voltage on the yield.

1.3.2. Pulse Trigger Flip Flop

In Pulse-triggered Flip-Flops (P-FF), definite types of sources are utilized for producing pulses. These consist of a strobe signal and a lock signal that manage data storage. When the change takes place in the signal edge, it generates brief clockwise impulses, which create a very small pulse width. This narrow pulse guarantees that the flip-flop will respond rapidly and correctly, providing precise operation. The Riegel action takes place at the flip-flop's cycle end for effective timing control.

Master - Slave configuration Two bars are used. P-FF complications are shortened by the master slave because there is only one lock bar required. There are also plenty of loans that exceed the clock cycle. These advantages do not only require the spread of the pulse clock to create a pulse generator circuit for a pulse width controller that is subtle in the development aspect of development.

Pulse generation techniques can be defined directly or indirectly by P-FFSIGNS. It is implicit of the P-FF type, and there is no external pulse generator circuit, and all external pulse signals are generated. In explicit type P-FF, PGs were not part of the design, and many classes of impulses from the FFS were expected to implement the adaptation of energy and delay, with subjects in submission structures (such as high-speed and low-energy).

Choosing the flip-flop used in the application is a very difficult task, as it depends on the size transmission of the transistor and already existing design. First, the right size and choice of methodology is important, and is important to achieve error-free results used in real designs. Flip-Flops are often used in computing circuits to operate at regular clock intervals for selected sequences next High Performance Flip Flop.

Flip-flops are habitually utilized in computational circuits to function in chosen arrangements amid repeating clock interim's.

Following are most high performance flip flops:-

1. Double Edge-Triggered Flip-Flop (DET)
2. True Single-Phase-Clock Flip-Flop (TSPC)
3. Clocked CMOS Flip-Flop (C2CMOS)
4. Single Edge-Triggered Flip-Flop (SET)

1.3.2.1. Double Edge-Triggered Flip-Flop

The input sampling is done both in the clock (double-edge triggered) and is done in the falling edge, the frequency of the clock can be called the flip flop starting FF double edge starting half of the clock frequency of the single edge. The double-edge trigger-flip-flop (DETFF) has less dynamic power consumption, constant power, and delay compared to previous flip flops. Since they have clock frequency semiconductor the double edge trigger failure is a popular technology of low energy designs. The single edge can be designed with two light-penetrating latches, parallel to a double edge trigger flip flop whereas single-edge triggered flip flop in series. Clock can be triggered by passing data on both positive and negative nodes as a flip-flopping double edge

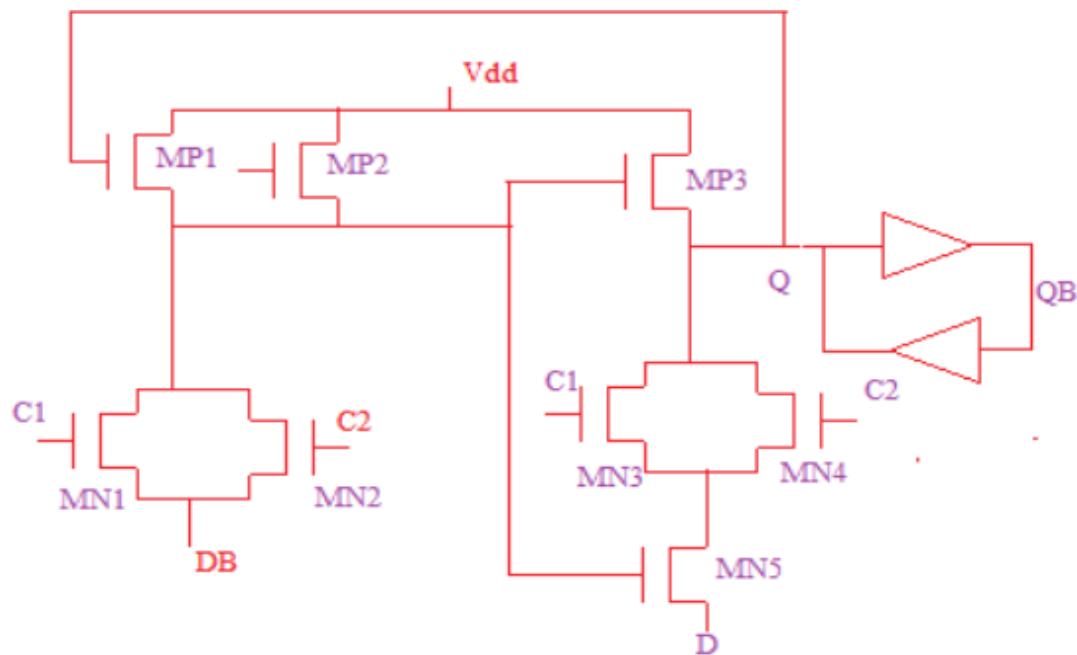


Fig 1.6: Double Edge-Triggered Flip-Flop (DET)

This outbuck supports many applications in different domains. Flip-flops are the key memory devices in digital circuits, where power and speed are the main issues. Another method of scaling makes use of double edges, with which the clock can react in both rising and falling edges. This method offers improved performance since it enables synchronization and data treatment in a superior manner. With the comparison of triggers that fire on single edges, circuits gain better depth and timing precision. Applying the same data to edge-to-edge comparison provides uniformity and efficiency in operations. In general, this method maximizes the performance and operation of digital systems in most applications.

1.3.2.2. True Single-Phase-Clock Flip-Flop

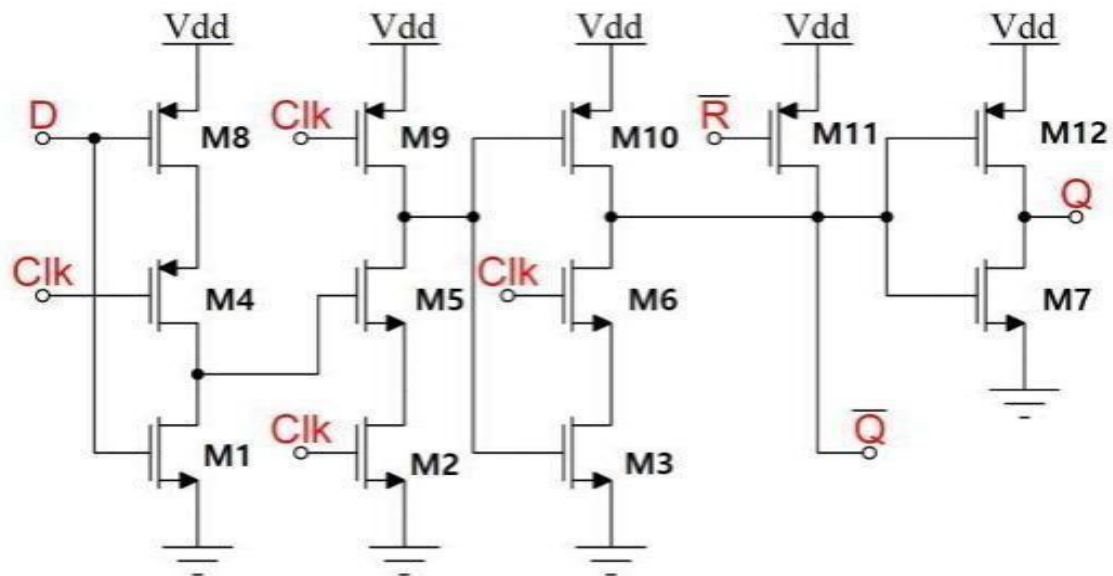


Fig 1.7: True Single Phase Clock (TSPC)

True Single-Phase Watch (TSPC) is a common dynamic flip-flop variety that performs fast, low-power flip-flop operations. TSPC Lock is a quick and simple construction that uses a single-phase watch. This lock was built by integrating two parts: CMOS Domino and CMOS Nora. Previously double-triggered flip-flops were designed in parallel using two complementary bars. The lock responded on the positive edge of the square clock pulse, and the other reacted to the negative edge.

The results of the two bars were combined into one result through a combination circuit that was optimized for signal integration. This circuit consists of a merge circle, where there are certain rules that dictate how outputs are computed. If one circuit shows a high-impedance (Hi-Z) state at the output node, it essentially isolates itself from affecting the result. In these instances, the other circuit assumes control, actively setting the output value. This alternating control guarantees that there is only a single source pulling the output at a time, so signal contention and data corruption are avoided. If the situation is reversed, the previously active circuit would go high- impedance, enabling the other to set its value. This technique is very handy in bus-based systems and common data lines, where there are multiple outputs which have to coexist without interference.

1.3.2.3. Clocked CMOS Flip-Flop

The basic circuit of the CMOS circuit is also the gate. First, we will examine the structure of a clock and its working principles. Just like with traditional CMOS gates, the output of the clock-CMOS gate must also be "restored". For traditional CMOS gates, the output is always clamped by either a conductive PMO transistor or soil with a conductive NMOS transistor with a low level output to the power supply. Therefore, the output of the BAR CMOS gate should also be "clamped" by switching to an electric clock, in order to rest "impulse". Because the power clock function changes, this MOS switch must be a complementary CMOS transmission.

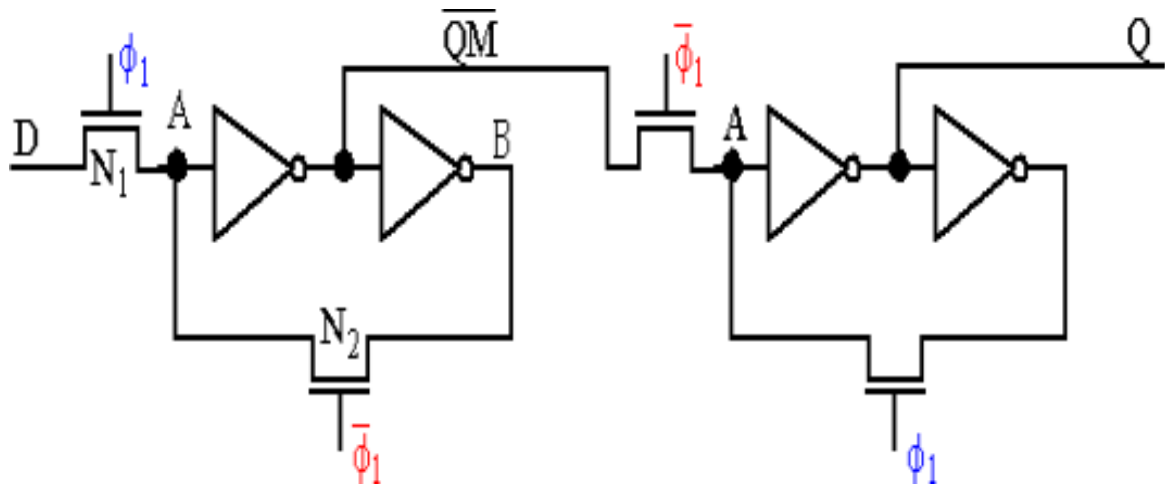


Fig 1.8: Clocked CMOS Flip flop

1.3.2.4. Single Edge Triggered CMOS

An edge-triggered simple flip-flop works by being triggered on a single clock edge, usually the rising edge. Data is latched only at the time that the clock signal crosses from low to high. Edge-triggered flip-flops are commonly found in digital circuits because they have high precision and control over timing. There are several types of edge-triggered flip-flops, including D, T, JK, and SR, with each being suitable for particular uses. For instance, D flip-flops are best for storing data, whereas JK flip-flops are best for toggling operations. These flip-flops are essential in applications such as counters, registers, and memory devices.

A single edge-triggered flip-flop operates by sampling the input at the exact moment when the clock transitions from one state to another. In a positive edge-triggered flip-flop, the state change occurs when the clock transitions from LOW (0) to HIGH (1).

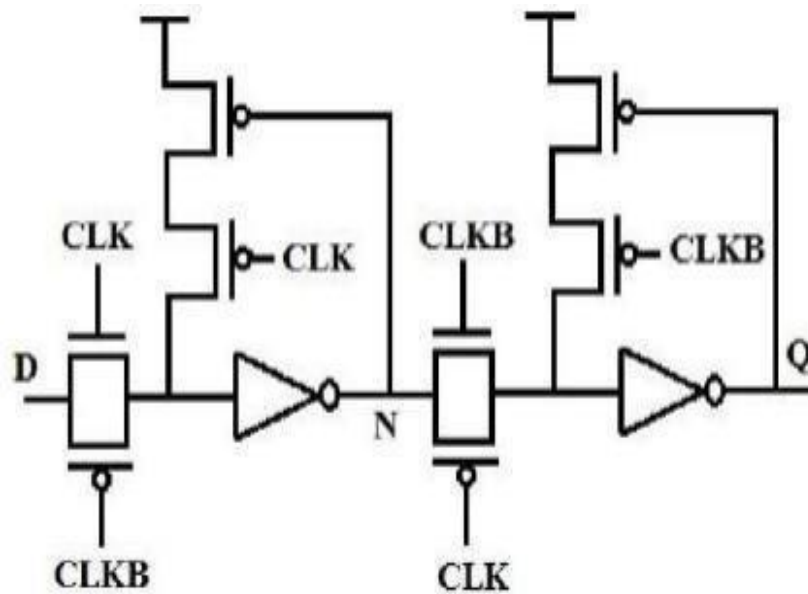


Fig 1.9: Single Edge Triggered CMOS Flip flop

In a negative edge- triggered flip-flop, the transition occurs when the clock changes from HIGH (1) to LOW (0).

CHAPTER 2

LITERATURE SURVEY

The evolution of Complementary Metal-Oxide-Semiconductor (CMOS) technology has played a pivotal role in the miniaturization of electronic devices, enabling the development of high-performance systems at increasingly smaller scales. However, as CMOS devices approach the nanometer scale, numerous challenges arise that affect the performance and reliability of hardware systems. The scaling of devices introduces several limitations, including issues related to power consumption, delay, noise, and variability. These factors significantly influence the performance of digital circuits, making it essential for engineers to optimize hardware components for higher efficiency and speed.

Among the most critical components in digital circuits are flip-flops—specifically D flip-flops (DFFs)—which serve as fundamental building blocks in Very-Large-Scale Integration (VLSI) circuits. DFFs are essential for storing and transferring binary data and form the foundation of sequential logic circuits. As the demand for high-speed processing increases, particularly in fields such as communication, computation, and medical imaging, the design of these flip-flops becomes a crucial focus for improving the overall performance of VLSI systems.

The design of D flip-flops has undergone significant advancements, particularly to reduce power consumption and improve operational speed. This paper explores the various types of DFFs and the metrics used to evaluate their performance. Additionally, it delves into the challenges and solutions related to DFFs, especially in high-frequency applications like Positron Emission Tomography (PET) scanners, and proposes new methodologies to enhance performance in terms of power delay product (PDP), energy-delay product (EDP), and overall area.

D Flip-Flops: Fundamental building blocks in VLSI circuits

A D flip-flop is a type of sequential logic circuit that captures the value of the data input (D) on the rising or falling edge of a clock signal. The DFF is one of the simplest and most widely used flip-flops due to its ability to store a single bit of information. The primary function of a D flip-flop is to store a bit of data and synchronize the movement of that data across different parts of the circuit, enabling sequential operations.

Types of D Flip-Flops: Active-edge D Flip-Flops: In active-edge DFFs, the data is latched based on the edge of the clock signal (either rising or falling).

TSPC (True Single Phase Clocking) D Flip-Flops:

TSPC DFFs are a more advanced type of DFF that are designed to operate with a single clock phase while reducing the power consumption compared to traditional designs. TSPC DFFs offer better performance in terms of speed and power efficiency, making them particularly suitable for high-frequency applications. In VLSI design, D flip-flops are often the bottleneck in achieving high-speed operations due to the inherent delay in their transition from one state to another. As CMOS technology continues to scale down to smaller nodes, such as 7nm or 5nm processes, the challenge becomes more pronounced due to the increased delay, reduced voltage margins, and higher power consumption. To address these issues, researchers have proposed various modifications to DFF designs, optimizing them for better performance metrics.

Performance Metrics for Flip-Flop Design:

Several key performance indicators (KPIs) are used to evaluate the effectiveness of D flip-flops in VLSI circuits. These metrics are crucial for comparing different DFF topologies and understanding their suitability for specific applications.

THE MAIN PERFORMANCE METRICS INCLUDE:

Power consumption: Power efficiency is one of the most important metrics in VLSI design, as the demand for low-power devices continues to grow. In high-frequency applications like PET scanners, where the system operates at MHz to GHz frequencies, reducing the power consumption of flip-flops is essential to avoid overheating and minimize the energy required for operation.

Propagation delay (d-q): The delay between the input (D) and the output (Q) of a flip-flop is a critical factor in determining the overall speed of a system. A shorter delay is desirable, as it allows for faster data processing and higher clock speeds.

Area: The physical area consumed by a flip-flop in the integrated circuit layout is another significant factor. Smaller area requirements allow for more compact designs and reduced costs. However, reducing area too much can lead to performance trade-offs, such as increased delay or power consumption.

Transistor Count: The number of transistors required for implementing a flip-flop impacts both the area and power consumption. A higher transistor count generally leads to increased power consumption and larger circuit area, which may not be desirable in resource-constrained

environments.

Number of Clocked Transistors: Clocked transistors refer to the transistors in the flip-flop that are driven by the clock signal. The fewer the clocked transistors, the lower the power consumption, but this must be balanced with other factors like reliability and performance.

Power Delay Product (PDP): PDP is a crucial metric that combines both power consumption and propagation delay. It provides a measure of the efficiency of a circuit in terms of both speed and power. Lower PDP values indicate a more efficient design that delivers high performance with minimal power usage.

Energy Delay Product (EDP): Similar to PDP, the EDP metric evaluates the trade-off between energy consumption and delay. A low EDP value signifies a good balance between speed and power consumption, making it a critical metric for energy-efficient designs.

Challenges in High-Speed Applications: PET Scanner Example. The application of flip-flops in specialized systems, such as Positron Emission Tomography (PET) scanners, exemplifies the need for high-performance flip-flops. PET scanners require complex signal processing to produce high-quality images, and the electronics used must operate at high speeds with minimal error rates. In this context, multichannel Analog-to-Digital Converters (ADCs) are essential components, and they rely on flip-flops to synchronize data transfer across the system. Traditional D flip-flops, especially passive ones, tend to struggle in high-frequency applications, such as those found in PET scanners, where clock speeds can reach several MHz. Passive D flip-flops often introduce delays that slow down the entire system, leading to performance degradation. To address this, TSPC D flip-flops are often selected due to their ability to operate at higher speeds with lower power consumption.

However, TSPC flip-flops are not without their challenges. They are susceptible to glitches in the middle of the clock cycle, which can result in timing errors and data corruption. These glitches arise due to the complexity of the circuit design, where multiple transistors and logic gates interact.

To mitigate the impact of these glitches and improve performance, researchers have explored various techniques such as clock gating, edge-triggered designs, and optimizing transistor sizing. Additionally, novel topologies have been proposed to minimize the area and power consumption of the flip-flop while maintaining high performance.

Proposed Low-Control, High-Performance Flip-Flop Design in response to the challenges of

traditional flip-flop designs, a new approach for low-control, high-performance flip-flops has been proposed. This new design aims to optimize the use of space, reduce power consumption, and enhance the overall performance of the flip-flop while minimizing glitches and other sources of error.

KEY FEATURES OF THE PROPOSED DESIGN INCLUDE:

Low Power Consumption: The design incorporates techniques such as dynamic voltage scaling and clock gating to reduce the power consumed during operation.

Reduced Area: By using fewer transistors and employing efficient routing techniques, the new design minimizes the area occupied by the flip-flop, making it ideal for highly integrated circuits.

Improved PDP and EDP: The design focuses on achieving a low Power Delay Product (PDP) and Energy Delay Product (EDP), ensuring that the flip-flop operates at high speeds with minimal energy usage.

Enhanced Clocking Mechanisms: The clocking scheme is optimized to reduce the chances of glitches and improve the synchronization of the flip-flop, especially in high-frequency applications.

CHAPTER 3

EXISTING METHOD

3. VARIOUS D-FLIP-FLOP ARCHITECTURES FOR TDCS

Advancement in semiconductor innovation has essentially upgraded the execution of computerized frameworks. In any case, control utilization remains a major challenge, especially in battery-operated and Web of Things (IoT) applications, where vitality productivity could be a best need. Flip-Flops are serving as quint essential establishments of computerized systems, contribute essentially to aggregate up control utilization.

Subsequently, optimizing their plan is vital for progressing vitality proficiency in cutting edge electronic frameworks. The D-flip-flop (D-FF) is one of the first broadly utilized sorts of FFs, serving as the essential component in successive circuits. It captures at the rising edge the clock, it records the input (D) and stores the esteem till the consequent clock cycle.

3.1 RIGHT SHIFTING VERNIER TDC

D-flip flops (D-FFs) are often employed to produce surrender signals in Vernier Time-to-Digital Converters (TDCs), which are necessary for precise time interval measurement. As depicted in Figure 1, both left-shifting and right-shifting Vernier TDCs are adapted for improved performance. In a left-shifting Vernier TDC, the negative edges of the Start and Stop signals progressively move towards the left, meaning earlier timing intervals. In contrast, in a right-shifting Vernier TDC, the positive edges of these signals are right-shifted, indicating delayed intervals. D-FFs ensure accurate edge detection, enabling dependable pulse alignment and enhanced time resolution in TDC operations.

The positive-edge-triggered (PET) DFFs within the right-shifting TDC and the negative-edge-triggered (NET) DFFs within the left-shifting TDC decide the exact minute when the quick flag adjusts with the moderate flag. By combining the Vernier TDC transistor for left wing and right shift on both delay lines, a bidirectional integrator for delaying time through both delay lines with much more, it achieves excellent accuracy as a standard delay line for two-way guests.

A standard edge-triggered D flip-flop is the most common type used in TDCs. It captures data on either the rising or falling edge of the clock and is commonly found in counter-based and delay-line-based TDCs.

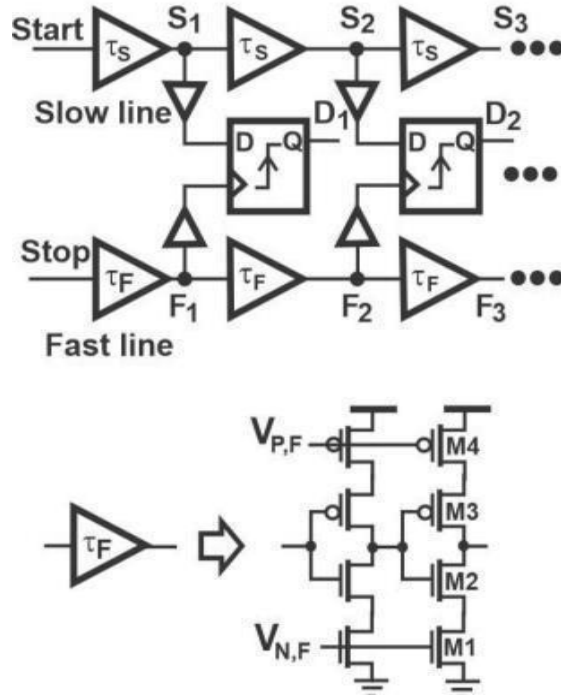


Fig 3.1: Vernier TDC with Right-Shifted Configuration

D-FF can be broadly isolated into three categories: sense amplifier (SA) DFF, energetic DFF, and inactive DFF. A normal inactive DFF is the transmission-gate (TG) master slave (MS) DFF shown in Figure 11. The positive edge triggered D-flip-flop guarantees that the yield moves as it were amid the rising edge of the clock, lessening glitches and making strides timing exactness. The master-slave D-flip-flop, comprising of two connected hooks, avoids race conditions by exchanging information in two stages to begin with to the ace hook and after that to the slave lock. Moreover, the Sense-Amplifier (SA) D flip-flop is optimized for control and speed, employing a sense intensifier for fast decision-making, making it well-suited for low-power and high speed applications.

A Vernier Time-to-Digital Converter (TDC) with right-shifting D-flip-flops (DFFs) measures time intervals by comparing the delays of two signals through a Vernier delay line, where the right-shifting DFFs are used to capture the output of the delay line stages.

In a right-shifting Vernier TDC, the output of the Vernier delay line is captured by a chain of DFFs, where the DFFs are clocked by a clock signal that is shifted to the right. This means that the output of the delay line is latched into the DFFs at successive clock edges, allowing for a finer time resolution.

3.2 TG MASTER-SLAVE D-FF

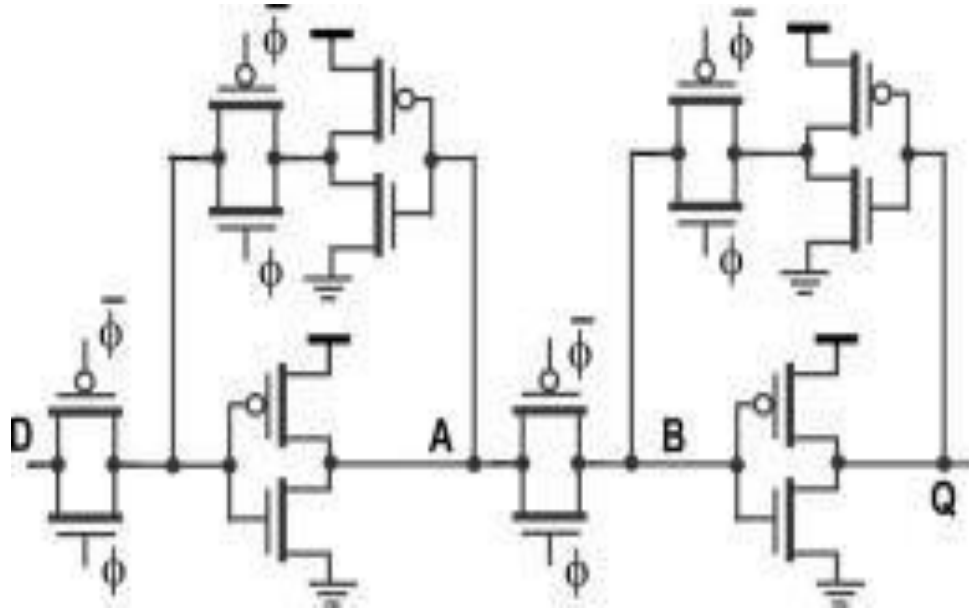


Fig 3.2: TG Master-Slave D-FF

A Transmission Gate Master Slave Flip-Flop is a flip-flop with a clock edge that uses an inverter and transmission grid to transmit and store data. It has two primary latches: a master latch and a slave latch. The Master Latch locks the input data with a high-level tact, and the slave latch controls the data to the decomposition point of the clock.

The heart of the transmission gate master-slave flip-flop is the transmission gate. A transmission gate is basically a switch constructed from a pair of complementary MOSFETs - one nMOS and one pMOS in parallel. With high clock signals, the gearbox is electrically conductive and data is passed. If the clock signal is low, the gearbox is not conductive and will cut the entry from the output.

A transmission gate master-slave ff works in two phases. In the high phase of the clock ($CLK = 1$), the transmission gate to the master latch is enabled, and the master latch captures the input data. The slave latch is disabled at this moment since the inverted clock signal ($CLK\bar{K}$) is low, disabling the transmission gate to the slave latch. When the watch moves to low phase ($clk = 0$), the master lock is turned off, but the gear gate attached to the slave block is switched on.

The heart of the transmission gate master-slave flip-flop is the transmission gate. A transmission gate is basically a switch constructed from a pair of complementary MOSFETs - one nMOS and one pMOS in parallel. The clock signal and its complement (CLK and $CLK\bar{K}$) control the gate. With high clock signals, the gearbox is electrically conductive and data is passed. If the clock signal is low, the gearbox is not conductive and will cut the entry from the output.

A transmission gate master-slave flip-flop works in two phases. In the high phase of the clock ($CLK = 1$), the transmission gate to the master latch is enabled, and the master latch captures the input data. The slave latch is disabled at this moment since the inverted clock signal ($CLK\bar{K}$) is low, disabling the transmission gate to the slave latch. When the watch moves to low phase ($clk = 0$), the master lock is turned off, but the gear gate attached to the slave block is switched on.

The master-slave flip-flop structure makes sure that the output is changed only on the falling edge of the clock pulse. This makes it impossible for glitches or noise to pass through directly from input to output, making the signal more stable. Because it is edge-triggered, the master-slave flip-flop is very well-suited for synchronous digital systems. Its capacity to maintain the output constant between clock edges makes it possible to store and transfer data dependably even at high operating speeds. Stability is important in today's high-speed digital circuits, in which proper timing and data integrity must be provided. Master-slave flip-flops, therefore, find extensive applications in processors, memories, and controllers.

The master-slave transmission gate flip-flop has many benefits. This allows data to be copied into the slave block of the master latch that represents the data in the output. It also minimizes the possibility of timing glitches as the output only gets updated at designated clock edges. Additionally, transmission gates used reduce leakage currents, enhancing power efficiency. The Master Latch locks the input data with a high-level tact, and the slave latch controls the data to the decomposition point of the clock.

This configuration guarantees that the output only changes at certain clock edges, so the flip-flop is immune to glitches and noise during the active phase of the clock. However, the design requires more transistors than a basic D flip-flop, increasing the complexity and silicon area of the circuit. Additionally, the flip-flop's performance can be affected by clock skew a mismatch between the clock signal and its complement which can lead to incorrect data storage.

3.3 POSITIVE-EDGE TRIGGERED D-FF

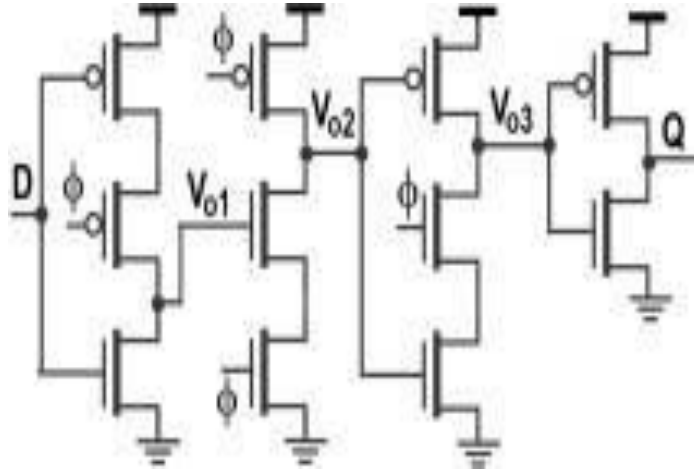


Fig 3.3: Positive-edge triggered D Flip-Flop

A Positive-Edge Triggered D Flip-Flop is a next-generation sequential circuit with high-speed and low-power characteristics ideal for use in contemporary digital systems. As compared to traditional D flip-flops, which use both clock (CLK) and its complement (CLK'), TSPC D-FF uses only one phase of the clock, which lowers clock skew, power consumption, and circuit complexity considerably. It is specifically helpful in high-frequency VLSI circuits since it can minimize dynamic power dissipation while ensuring stable operation. It is constructed based on dynamic logic principles, and there are two essential stages: the master stage, which holds the input (D) when the clock is high, and the slave stage, through which the held value is passed to the output (Q) on the positive edge of the clock. This structure obviates the necessity of additional latches or clocked inverters and is more efficient than traditional flip-flops.

As TSPC flip-flops are transit efficient, they consume less chip area and power, which makes them suitable for low-power microprocessors, DSP processors, and high-speed ASICs. Moreover, TSPC D-FFs are broadly realized using CMOS technology, wherein NMOS as well as PMOS logic styles can be utilized to attain optimal speed and power trade-offs. Positive-edge triggered True Single-Phase Clocked (TSPC) D Flip-Flops find extensive applications in numerous high-speed and low-power applications. In CPUs and microprocessors, they are employed as pipeline registers to facilitate efficient data transfer between processing stages with minimal power consumption.

Likewise, in digital signal processing (DSP) blocks, TSPC D-FFs are employed in high-speed data registers to process audio, video, and communication signals in real time while minimizing clock load. These flip-flops are also fundamental to low-power VLSI circuits and thus are suitable for battery-powered applications like smartphones, IoT devices, and wearables, where power consumption needs to be minimized. TSPC D-FFs are also important in clocked storage elements like register banks and memory elements, where data storage needs to be stable and power consumption should be efficient. They are also critical in high-speed communication networks, utilized in serial-to-parallel and parallel-to-sequential converters for data transmission and clock distribution networks for synchronized data transfer in optical and RF communication. Additionally, in ASIC (Application-Specific Integrated Circuits), TSPC D-FFs are embedded in custom-built chips for AI processors, automotive electronics, and medical devices where high efficiency and low power are essential. Their effectiveness in GHz-frequency applications renders them a crucial building block in contemporary integrated circuit design.

3.4 SENSE AMPLIFIER D-FF

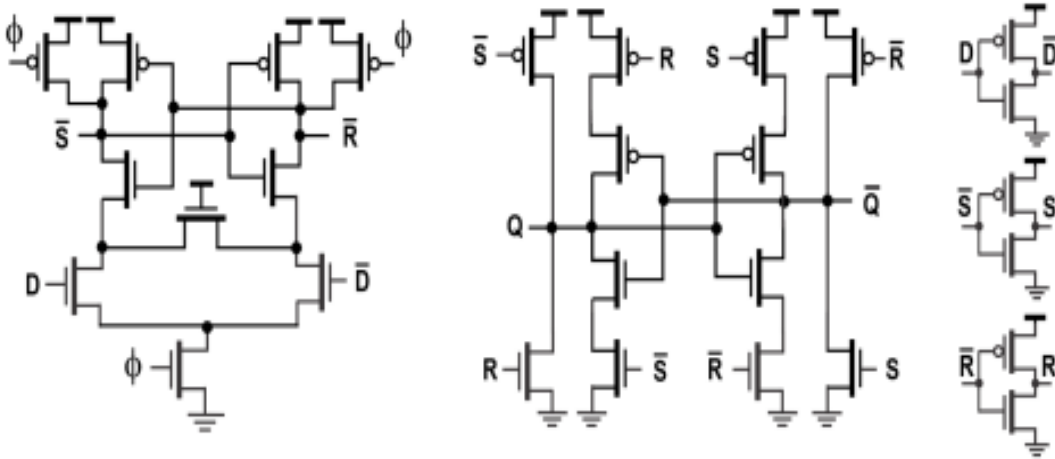


Fig 3.4: Sense Amplifier D-FF

A Sense-Amplifier D Flip-Flop (SA D-FF) is a dedicated high-speed flip-flop that has the capability to offer low latency, high speed, and low power dissipation. It is extensively applied in contemporary high-speed microprocessors, high-speed VLSI circuits, and clock distribution networks. As opposed to traditional D flip-flops that are based on several clocked stages to latch the data, the SA D-FF is based on differential sensing and amplification and is therefore more efficient and faster.

The SA D-FF has two primary functional blocks: a sense-amplifier stage and a latching stage. The sense-amplifier stage is used to sense small voltage differences between differential input signals and amplify them quickly. This characteristic is especially beneficial in low-power and high-speed applications, where signal integrity has to be preserved even with decreased voltage swings. After the input data is detected, the latching stage stabilizes the output and retains it until the subsequent clock cycle. SA D-FF is designed to operate effectively at high frequencies while drawing less power than traditional master-slave flip-flops. The SA D-FF's primary strength is its reduced latency and increased clock-to-Q propagation time, making it greatly appropriate for pipelined as well as high-speed computing implementations. In addition, dynamic power consumption decreases through the suppression of redundant switching, thus, making it quite appropriate for VLSI energy-efficient designs.

The reason the sense-amplifier stage is differential for inputs is so that it could increase signal immunity to noise since it is implemented in low-voltage and also high-speed digital circuits. It is another significant advantage that it saves chip area consumption without compromising the superior performance it offers. Sense-Amplifier D Flip-Flops are, in fact, utilized in microprocessors with very high-performance requirements to enhance the speed of processing data and limit pipeline delays. They are widely used in the design of memories, especially for SRAM and cache memory applications, where very fast sensing and latching of data are of prime importance in overall system performance. In high-speed communication circuits, SA D-FFs provide reliable and noise-immune data capture and hence are beneficial in clocked storage elements, frequency synthesizers, and data converters.

They are also widely applied in clock distribution networks, where their low power and high-speed properties assist in maintaining signal integrity over large-scale integrated circuits. Overall, SA D-FFs are a vital element of current high-speed digital circuits that make possible increased speed of computation, reduced power, and increased signal integrity. Due to the differential sensing approach together with the operation at high speed, they find preference where high efficiency coupled with very minimal timing delay is necessary.

CHAPTER 4

PROPOSED METHOD

4.1 STRUCTURE OF TGFF

The transmission flip-flop (TGFF) is a sequential circuit commonly used in digital electronics. It uses transmission gates to effectively control data flow and improve performance. The TGFF structure usually consists of two locks, one acts as a master and one slave, synchronized by additional watch signals. During the positive clock, the main course obtained input data while the slave lock contained the previous state. Conversely, when the clock is low, the slave lock will update its condition to correspond to the owner's arrest data. Gate play an important role in the TGFF because they allows effective signal conversion with minimum spread delay. Unlike conventional concepts based on passing semiconductor balls, transmission doors use both NMO and PMOS semiconductors, improving the integrity of the signal and reducing stress reduction problems. This combination leads to improvement in noise rates and better performance in low -power applications.

TGFF is especially popular in the VLSI design because of their compact structure and the number of semiconductor balls decreased compared to traditional seesaw concepts. Their effective operation makes them ideal for high -speed digital circuits, watch distribution networks and low -power applications. In addition, their ability to minimize clock deviation helps improve synchronization accuracy in synchronous circuits. Therefore, TGFF is widely applied in modern semiconductor notions, in which energy efficiency and performance optimization are essential.

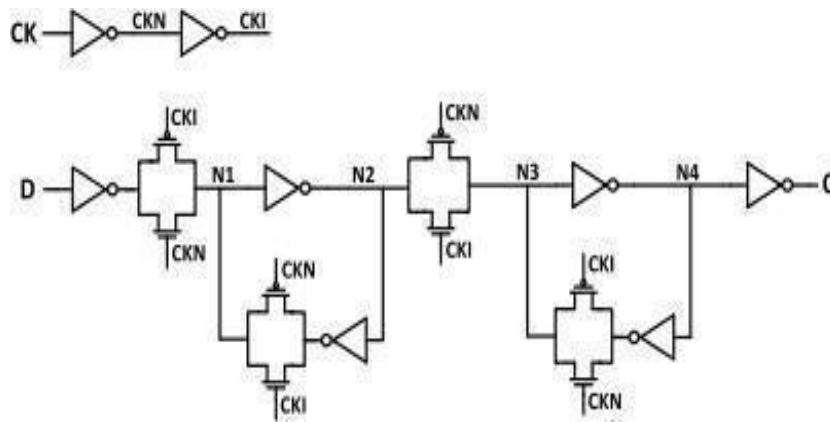


Fig 4.1: Schematic Design of the TGFF.

The tilt of the transmission gate (TGFF) is the most used FF in contemporary digital systems. A FF is not interrupted in well -operating in situations near the threshold of TGFF. The large clock network is the main defect in TGFF. In addition to controlling a larger number of semiconductor balls, the interior buttons of CKN and CKI change any input data. Therefore, even when the data of the data is still low, the power consumption of TGFF is always high. Optimizing the use of additional clock signals is necessary to reduce energy consumption to of FF. In previous publications, many single and low -power FFS and time were presented but there are still some issues that have an impact on the consumption of these FFs. For example, some FFS suffer from high before high power and other functional disorders at low supply voltage. The article shows that the open power with the block is (TSPC) to solve these problems. The proposed FF can operate at a wide supply voltage and not discord. In addition, FF is proposed to completely eliminate the previous backup load and significantly optimized the energy consumption compared to the FF of the low front capacity.

4.2 STRUCTURE OF TSPC FF

TSPC Logic (True Single - Phase Clock) is a wide range of design technology for digital circuits, especially high speed and low power applications. The TSPC circuit uses a single-phase clock signal to control data flow, eliminating the need for a complementary clock signal. This design simplifies clock distribution, reduces power consumption, and minimizes timing issues such as clock bales. TSPC structures are typically based on a combination of dynamic logic and locking elements, allowing for faster switching speeds and improved performance. The TSPC circuitry operates efficiently in one clock phase, reducing the complexity of the tux trout, making it suitable for large-scale integration of VLSI designs.

The True Single-Phase Clock (TSPC) flip-flop records data and latches it on a dynamic memory node, speeding up performance and minimizing the use of transistors over standard flip-flop implementations. The reduced architecture avoids parasitic capacitance, adding to circuit speed and energy efficiency. TSPC flip-flops use just one clock phase, making them easier to handle timing specifications and ideal for high-speed and low-power applications. This makes them well-suited for use in contemporary digital systems like processors and high-performance computing systems, where higher data processing speed and reduced power consumption are necessary to improve system efficiency and reliability overall.

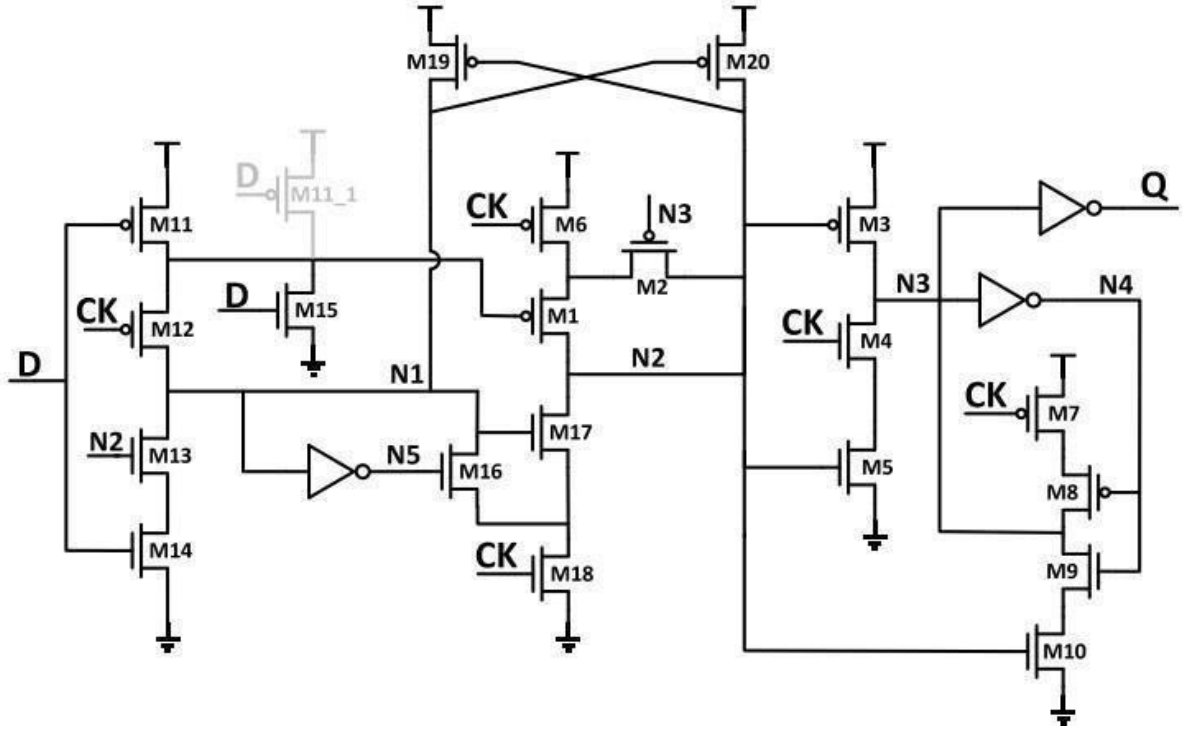


Fig 4.2: Schematic Design of the proposed TSPC FF.

Since the voltage of floating nodes may change after transition due to the leakage current, it is necessary to carefully analyze floating nodes to avoid the generation of short-circuit paths. With the insertion of the input-aware transistor, the voltage of the node N2 is no longer pre-charged to VDD at the negative half cycle of CK when the input is 0. So the status of the node N2 needs to be analyzed carefully.

When the output $Q = 1$, which means $N3 = 0$, the state of N3 is maintained by M9 and M10. To keep the transistor M3 OFF and the transistor M10 ON at that time, the voltage of N2 needs to keep high. In order to avoid N2 becoming floating while the next input data is 0, a transistor M2 is inserted to keep the voltage of N2. M2 is controlled by N3 and provides a pre-charge path for N2 when $Q = 1$.

When the output $Q = 0$, $N3 = 1$, the state of N3 is maintained by M7 and M8. If the Pre-charge path of N2 is cutoff at that time, which means the input data is 0 and the voltage of N2 does not charge to high when CK is low, the only effect is that M3 is ON, which will not lead to a short circuit path.

When $N3 = 1$, $CK = 0$, and $D = 0$, the node N3 is isolated from its pull-down path (M4 and M5) through M4 ($CK = 0$), so the voltage of N2 has no effect on the node N3. Similarly, N1 is isolated from its pull-down path (M13 and M14) through M14 ($D = 0$), and the voltage of N2 has no effect on the node N1. Therefore, the floating of the node N2 is negligible in this case.

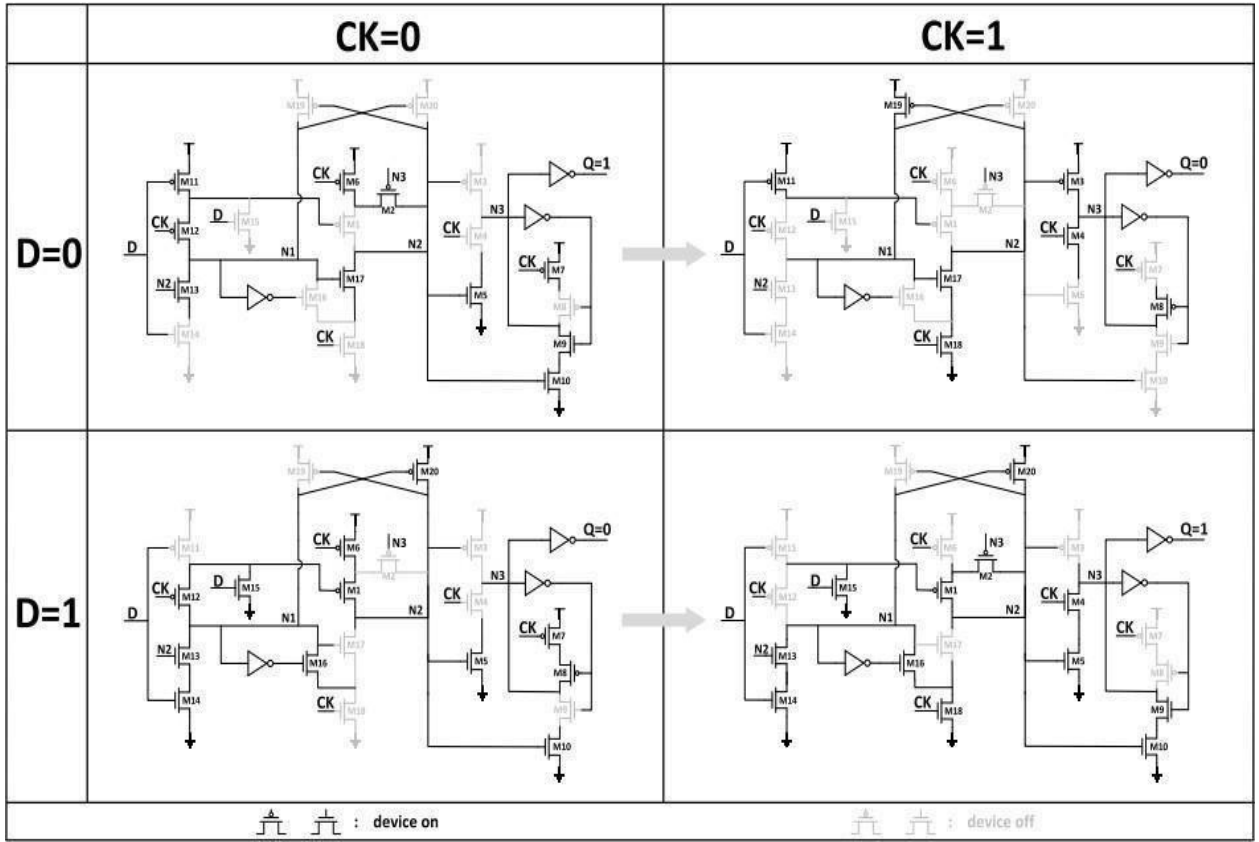


Fig. 4.3: Operation diagram of the proposed FF at different D and CK.

Transition from High to Low: When CK is low, node N1 charges to VDD via M11 and M12, node N2 charges to VDD via M6 and M2, node N3 remains low via M9 and M10, and output Q remains high. N2 is discharged into GND via M17 and M18 near the rising edge of CK and M13 is subsequently cut off to separate the FF from modifications in the input data. When the node N3 is concurrently charged to VDD via M3, the output Q drops to zero. The voltage of N1 stays high through M19 during the positive half cycle of CK, while the voltage of N2 stays low through M17 and M18.

Low-to-High Transition: N3 remains high through M7 and M8, the output remains low, N2 is charged to VDD via M6 and M1, and N1 is discharged to GND through M13 and M14 while CK is low. The output Q turns to 1 at the rising edge of CK when the input data is separated by M12 and the node N3 is discharged to GND via M4 and M5. The voltage of N1 stays low through M16 and M18 throughout the positive half cycle of CK, while the voltage of N2 stays high until M20.

4.3 SCHEMATIC DESIGN OF THE PROPOSED FF WITH SET

In digital electronics, the term "set" refers to a condition in which a particular logic device or memory element is forced into a defined state, usually representing a binary "1." This process is very important for continuous circuits such as flip-flops and bars, with defined functions sets the output to a high logic level. The specified conditions are often triggered by a dedicated input signal called a "set" or "preset" pin. If this signal is active, the output is driven with the desired logic "1" status, regardless of other inputs. This function is usually used to initialize circuits and to ensure specific output conditions for various digital systems. The defined functions play a key role in control circuits, meters, and storage devices that require accurate state control. In some structures, the set entrance may operate asynchronously.

This means that you can overwrite the clock signal and quickly put the output into high conditions. Importance of this feature is extremely important for scenarios that require rapid response or emergency conditions. The combination of set and reset functions gives you better control over the status of your system. This ensures stable and predictable operation with complex digital logic circuits.

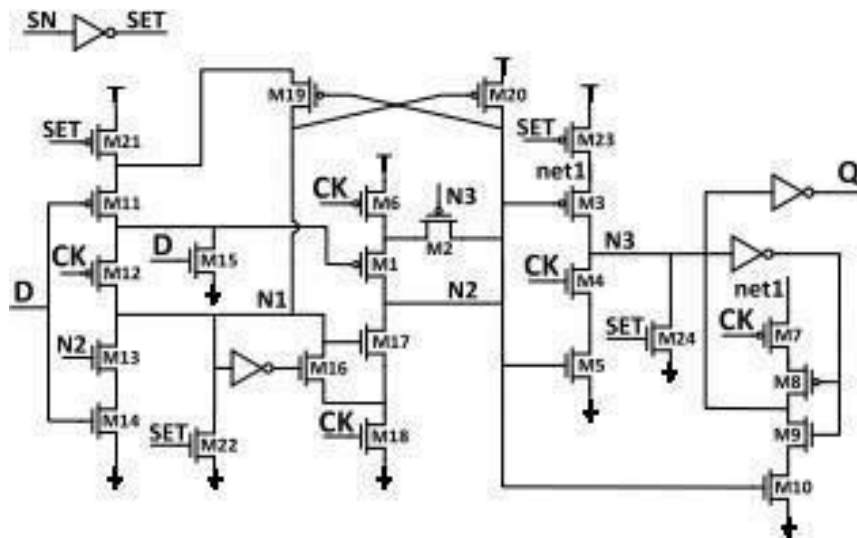


Fig 4.4: Schematic Design of the proposed TSPC FF with Set

N1's charging path is blocked off by M21 and N1 is dragged down through M22 when the set signal SN is low and SET is high. N2 is charged to VDD via M20 as node N1 is low. At the same time, M23 cuts off N3's charging path, pulling N3 downward through M24 while maintaining a high output.

As the transient waveform of the proposed FF with a set function given below, the output Q stays high when SN is low, the FF is in the set state. On the other hand, when SN is high, the FF operates normally.

4.4 SCHEMATIC DESIGN OF THE PROPOSED FF WITH RESET

In digital electronics, the term "reset" refers to a process that forces a circuit or memory element in a predefined state that normally represents a binary "0". This operation is essentially important to ensure that known conditions are assumed for continuous circuits such as flip-flops, registers, and meters. The reset function is usually activated by a dedicated input spin called a "reset" or "delete" signal. If this signal is asserted, the output is driven to a lower logic level regardless of the other inputs. This ability is extremely important for digital system initialization and ensures that it starts in a stable and predictable state, depending on the power supply or error condition.

The reset function can function synchronously or asynchronously depending on the circuit design. In a synchronous reset system, the reset signal is used in conjunction with the clock signal to ensure control timing. The reset function is extremely important for applications where system stability and proper initialization are important.

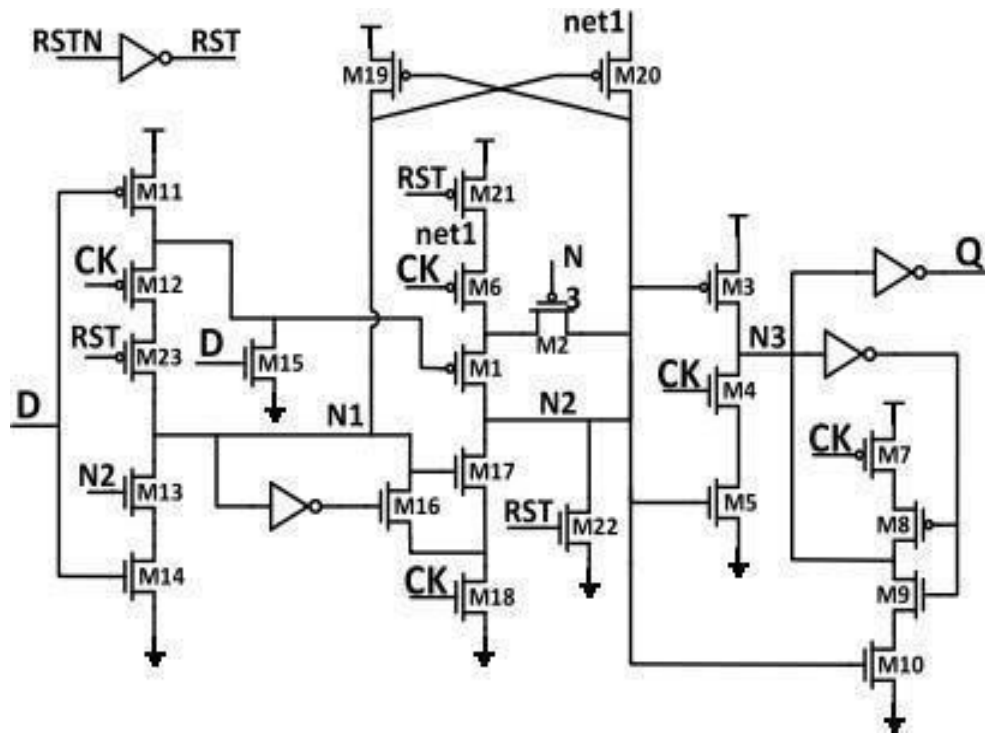


Fig 4.5: Schematic Design of the proposed FF with reset.

The proposed FF's schematic with the set feature is shown in the above figure. When the reset signal is received, M21 cuts off N2's charging path, and M22 pulls N2 downward. RST is high and RSTN is low. Because N2's low level causes N3 to be overly charged through M3, the output Q stays low. At the same time, M19 charges N1 excessively. When $CK = 0$ and $D = 1$, a PMOS M23 is added to isolate N1 from the input and stop short circuit current flowing through M12 and M15. When the RSTN is low, the output Q stays low, signifying that the FF is in the set state. On the other hand, when the RSTN is high, the FF reliably captures the input data at the rising edge of CK.

4.5 SOFT-ERROR TOLERANT OPTIMIZATION OF THE PROPOSED FF

Soft-Error Tolerant Optimization of the Proposed FF. Forceful voltage scaling will make computerized frameworks more helpless to delicate blunders, indeed whereas bringing down the supply voltage can significantly decrease control utilization. The proposed FF can be solidified to resist the single occasion impact (SEE), which incorporates single occasion disturbed (SEU) and single event temporal (SET), to extend the steadiness of low-voltage frameworks.

The temporal waveform of a SEU at hub N2 in both the proposed FF and TGFF. Like TGFF, the hub N2 within the proposed FF is helpless to SEUs. This is often since the information put away within the FF is eradicated when the voltage of N2 diminishes from VDD to GND as a result of a SEU, M3 and M19 are ON, and the hubs N3 and N1 are charged to VDD. Likewise, SEUs can influence hubs N1, N3, N4, and N5 within the recommended FF. It has been illustrated that C-element-based rationale may be a fruitful solidifying procedure. The proposed FF can be heartless to SEUs by utilizing rationale based on C components.

The proposed SEU-tolerant FF's helplessness to SEUs is inspected utilizing hub N2 as case. Since hubs N1 and N3 are isolated from N2 by the transistors controlled by N2R, hubs N1 and N3 within the proposed SEU-tolerant FF will not charge to VDD when hub N2 drops from VDD to GND due to a SEU.

A Soft Error Upset (SEU) is a type of error in digital electronics that occurs when a charged particle or radiation strikes a semiconductor device, causing a temporary change in its data state. Unlike permanent hardware failures, SEUs are non-destructive and result in transient faults that can disrupt data storage or logic states. Single Event Upsets (SEUs) are a class of errors typically occurring in memory components such as SRAM and flip-flops, in which an unexpected change in a stored bit from '0' to '1' or '1' to '0' due to a high-energy particle hit may result.

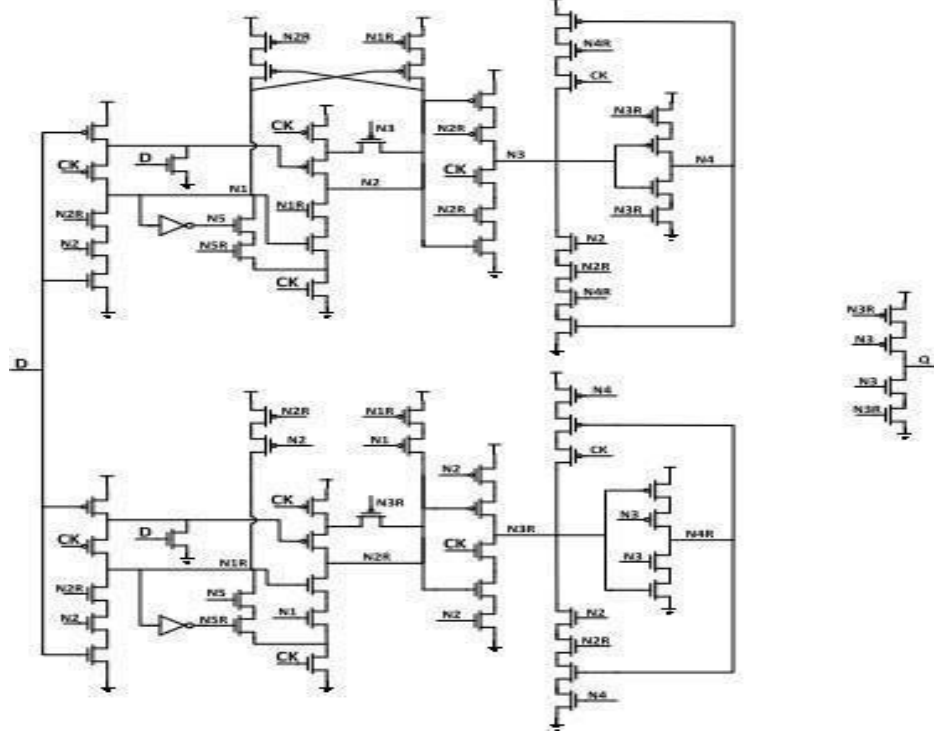


Fig 4.6: Schematic of SEU

Bit flipping takes place without any resultant long-term harm to the hardware but may manifest as incorrect data processing, system failure, or even failure in crucial operations. SEUs are a key issue in highly exposed radiation environments, like aerospace systems, satellites, and spacecraft, where cosmic rays or solar particles can readily couple with electronic devices. Even in cutting-edge computing systems on Earth, particularly in high-end or mission-critical systems, SEUs may happen and undermine system integrity. To overcome this problem, designers frequently use error detection and correction mechanisms to ensure data accuracy and increase reliability.

The node N2 in the proposed FF is susceptible to SEUs, similar to TGFF. This is because when the voltage of N2 drops from VDD to GND due to an SEU, M3 and M19 are ON, the nodes N3 and N1 are charged to VDD, and the data stored in the FF is overwritten. Similarly, the nodes N1, N3, N4, and N5 in the proposed FF are all susceptible to SEUs.

The node N2 is taken as an example to analyze the susceptibility of the proposed SEU-tolerant FF to SEUs. When the node N2 in the proposed SEU-tolerant FF drops from VDD to GND due to an SEU, nodes N1 and N3 will not charge to VDD since nodes N1 and N3 are isolated from N2 by the transistors controlled by N2R.

CHAPTER 5

TANNER EDA

The tool used for simulation purpose for the entire research work is Tanner EDA tool version 13.0 and micro wind for layout designing the features and functionality of this tool has been described below:

5.1 SIMULATION TOOL

The cycle of designing for enhancement of electronic circuits features a pre-fabrication verification step as an integral part. For a successful design, since fabrication costs are very high and it is time-consuming, proper verification is required. In this stage, Electronic Design Automation (EDA) tools serve an important purpose in that they assist designers in circuit functionality designing and verification. These tools verify and solve the differential equations characterizing the circuit behavior and output simulation results. This provides designers with the ability to simulate, verify, and optimize their circuits prior to actual production, minimizing mistakes and conserving resources. Proper utilization of EDA tools guarantees optimal performance and reliability in the end product.

Tanner EDA apparatus may be a total circuit plan and examination system that includes:

Schematic Editor (S-Edit): Schematic editor may be a effective plan capture and examination bundle that can create netlist specifically usable in T-Spice reenactments.

T-Spice Circuit Test system: T-Spice performs quick and exact reenactment of analog and blended analog/digital circuits. The test system incorporates the most recent and best gadget models accessible, as well as coupled line models and back for client- defined device models through tables or C capacities. All of SPICE's gadget models are joined, as well as resistors, capacitors, inductors, common inductors, single and coupled transmission lines, current sources, voltage sources, controlled sources, and a full complement of the most recent progressed semiconductor gadget models from Berkeley and Philips Labs.

Waveform Editor (W-Edit): W-Edit shows T-Spice reenactment yield waveforms as they are being produced amid reenactment. W-Edit may be a waveform watcher that gives ease of utilize, power, and speed in a adaptable environment planned for graphical information introduction.

Layout Editor (L-Edit): Leather treater EDA instrument incorporates L-Edit for format altering, Intuitively DRC for real-time plan run the show checking amid altering, Standard DRC for various leveled DRC, Standard Extricate for netlist extraction, Standard LVS for format versus schematic, Hub Highlighting for highlighting all geometry related with a hub and SPR for standard cell put &course.

5.2.T-SPICE

To convert your thoughts into plans, you must be able to mimic expansive circuits rapidly and with a high degree of accuracy. Meaning you wish a recreation device that provides quick run times, coordinating along with your other plan instruments, and is congruous with industry measures. Leather treater T-Spice Circuit Test system puts you in control of recreation employments with an easy-to-use graphical interface and a speedier, more instinctive plan environment. With key highlights such as multi-threading back, gadget state plotting, real-time waveform seeing and examination, and a command wizard for less complex Zest sentence structure creation, T-Spice spares you time and cash amid the recreation stage of your plan stream. T-Spice empowers more precise reenactments by supporting the most recent transistor models—including BSIM4 and the Penn State Philips (PSP) demonstrate. Given that T-Spice is consistent with a wide extend of plan arrangements and runs on Windows and Linux stages, it fits effectively and taken a toll viably into your current device stream.

T-Spice joins various advancements and changes not found in other Flavor and SPICE-compatible test systems:

- **Speed:** T-Spice gives profoundly optimized code for assessing gadget models, defining the frameworks of straight conditions, and understanding those frameworks. In expansion to the standard coordinate show assessment, T-Spice moreover gives the alternative of table-base transistor show assessment, in which the comes about of gadget show evaluations are stored in tables and reused
- **Convergence:** T-Spice employs state-of-the-art math methods for utmost numerical stability. It has the ability to analyze large, sophisticated feedback circuits which are troublesome for other SPICE packages. Because of this, T-Spice is an effective solution to precisely model and verify complicated electronic designs while working on developing and testing a circuit.
- **Accuracy:** T-Spice employments exceptionally exact numerical strategies and charge preservation to realize predominant reenactment exactness.
- **Macro displaying:** T-Spice reenacts circuits containing “black box” large scale gadgets.

A large scale gadget can straightforwardly utilize exploratory information as its gadget show.

- Macro gadgets can moreover speak to complex gadgets, such as rationale entryways, for which as it were the generally exchange characteristics, are of intrigued.
- **Input language extensions:** The T-Spice input dialect is an enhanced form of the standard Zest dialect. It contains numerous improvements, counting parameters, arithmetical expressions, and a capable bit and transport input wave determination language structure.
- **External model interface:** You'll be able create custom gadget models utilizing C.
- **Runtime waveform seeing:** The W-Edit waveform watcher shows graphical comes about amid reenactment. T-Spice examination comes about for voltages, streams, charges, and control can be composed to single or different records. Let's see how to plan for a CMOS inverter utilizing leather treater apparatuses.

STEP 1

Opening S-edit platform:

First of all, double click on the icon of s-edit on the desktop (or)

Go to the start menu >>All Programs >Tanner EDA >Tanner Tool v 13.0 > S-Edit v13.0



Fig 5.1: Opening s-edit platform

STEP 2

A new window will open:

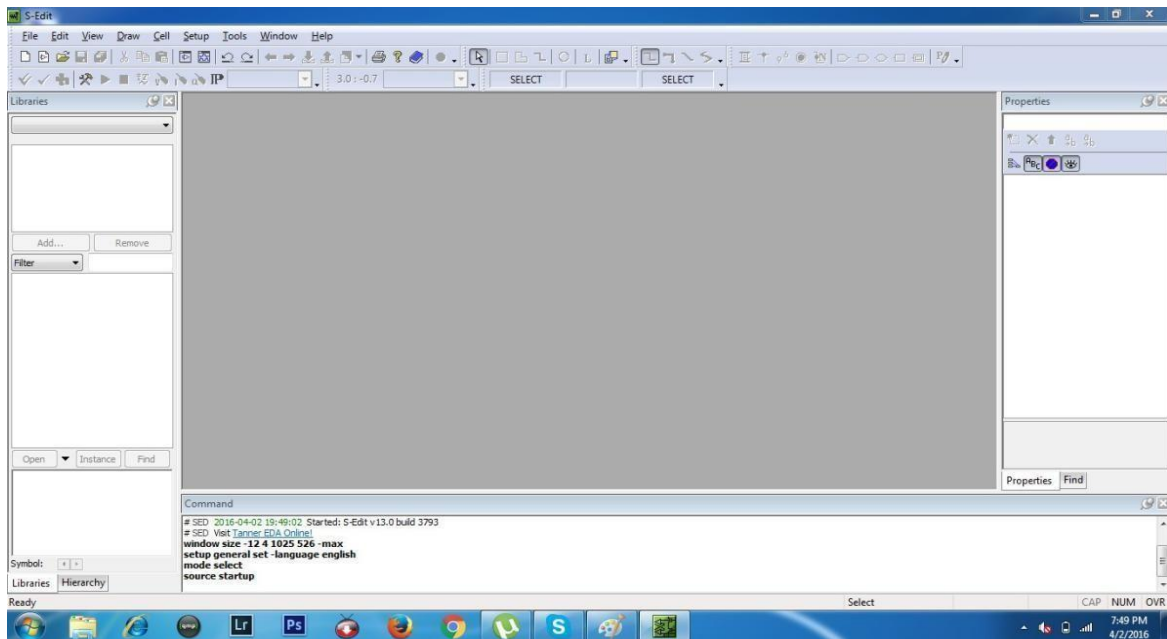


Fig 5.2: Opening Window

STEP 3

Go to >>file >> New >> New Design

Select New Design

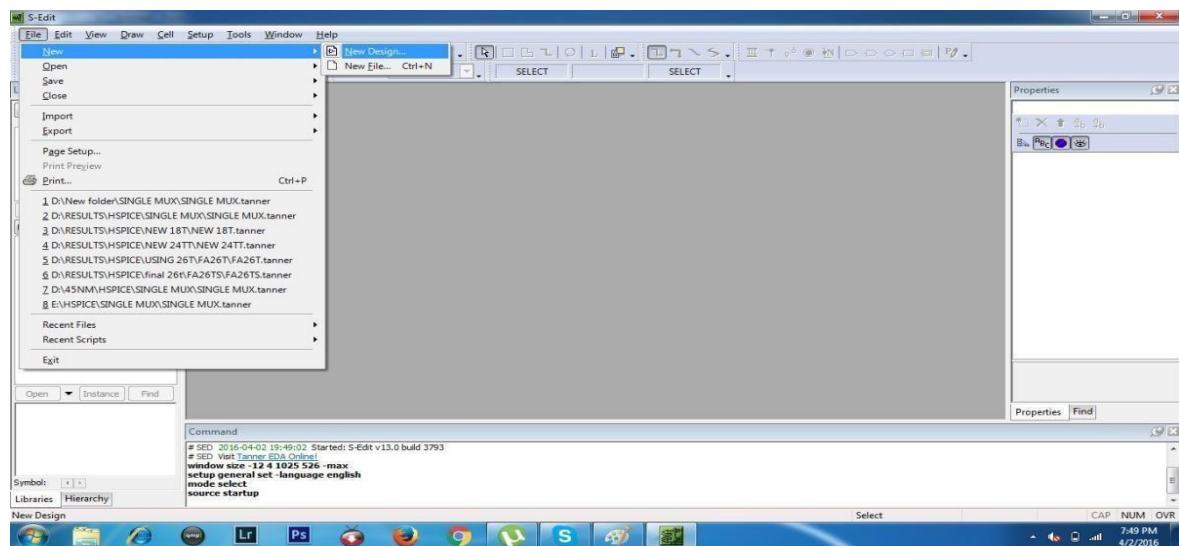


Fig 5.3: Creating new design

STEP 4

One dialog box will appear

Design Name : Give the name your design as you wish

Create a Folder : Give the path where you want to save the S-Edit Files.

Click OK.

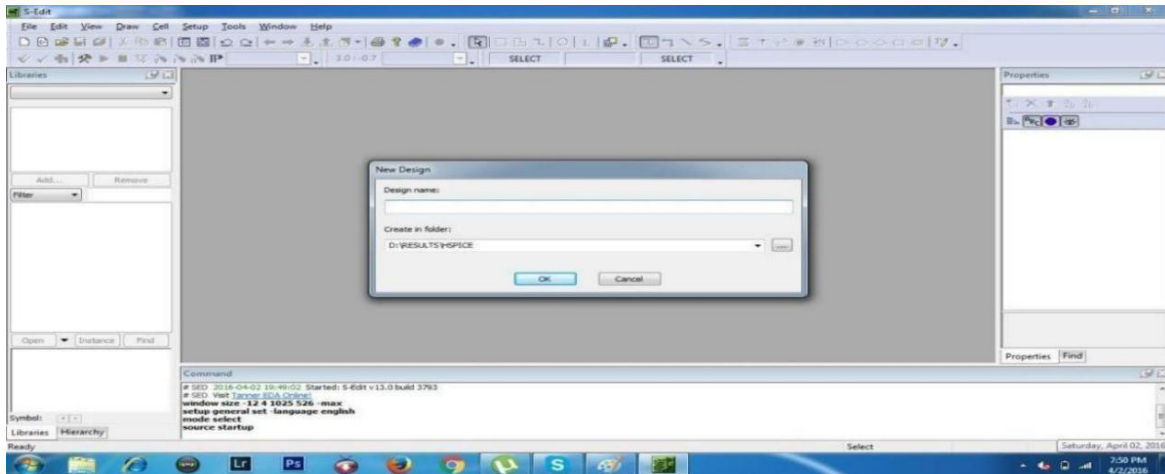


Fig 5.4: Naming the design

STEP 5

Now to add libraries in your work click on Add, left on the library window. Give the path where Libraries are stored C:\Documents and Settings \My Documents \Tanner EDA \Tanner Tools v13.0 \Libraries \All \All. Tanner

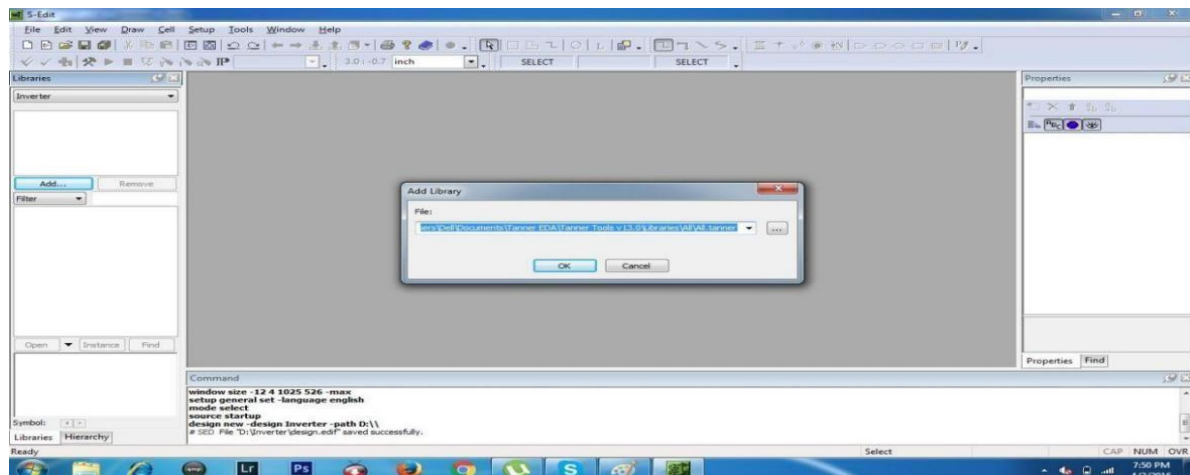


Fig 5.5: Adding libraries files

STEP 6

Now go to cell menu->>Select New View

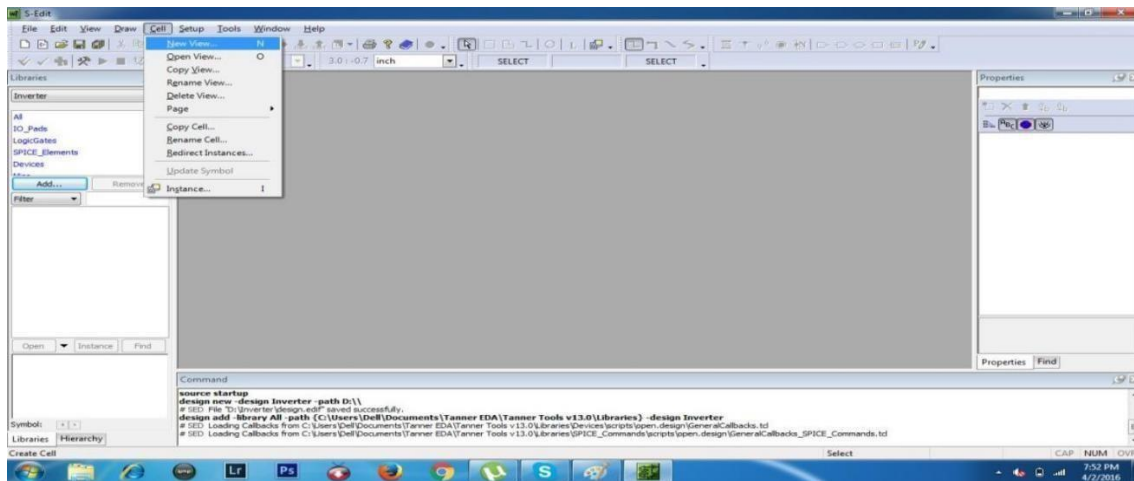


Fig 5.6: Creating new view

STEP 7

The new cell will appear like below:

Design = Your design name

Cell = cell no. (cell no you can change but your design name inverter will be same for different cell.

(Design name should be changed only when you are going to design another circuit)

View type = Schematic Interface name = by default

View name = by default Then press OK.

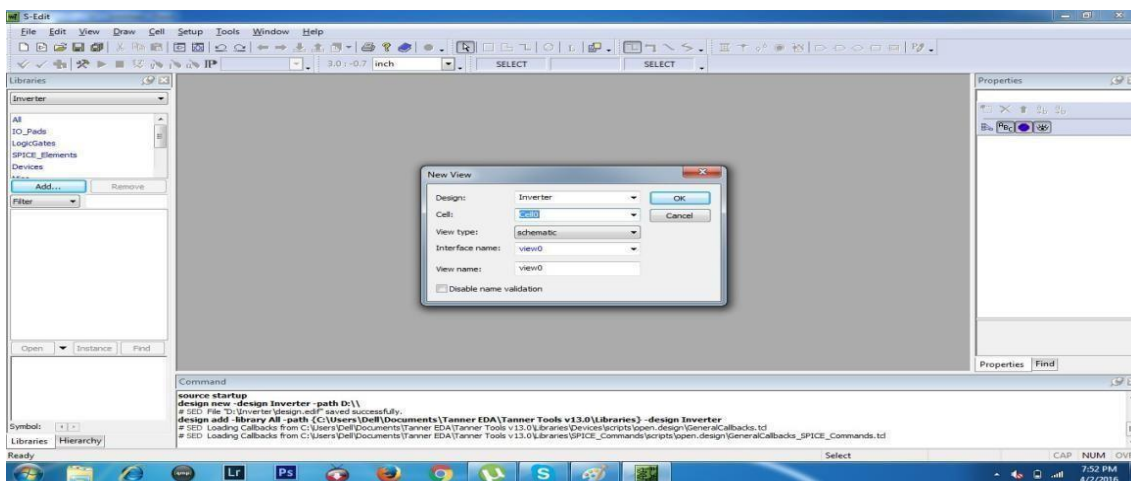


Fig 5.7: Naming the cell

STEP 8

Then a cell will be appeared where we can draw the schematic of any circuit. In the black window you have seen some white bubble arranged in specific order. This is called grid. You can change grid distance by clicking on black screen and scroll the mouse

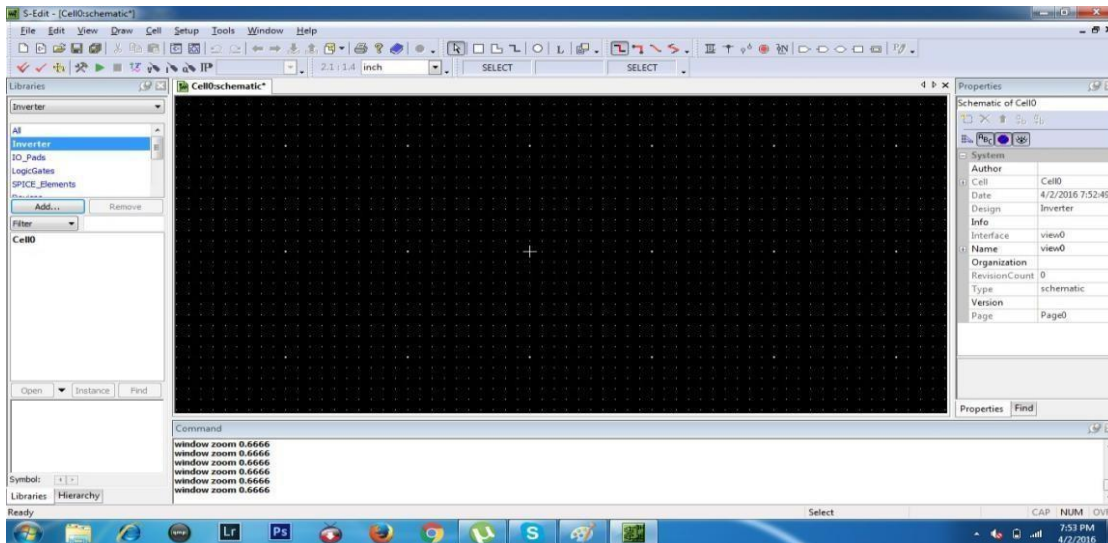


Fig 5.8: Opening the cell

STEP 9

Select any device from the libraries for example we have taken NMOS

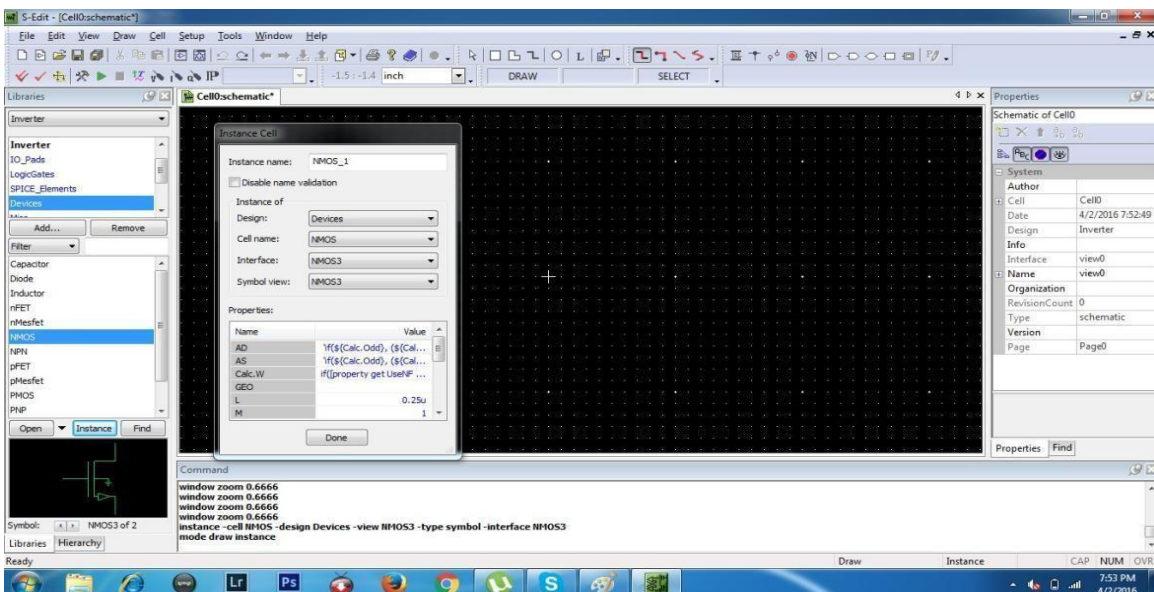


Fig 5.9: Selecting the element

STEP 10

- You can change the values of various device parameters according to your requirement.
- Go to properties >> change the parameter values as your requirement.
- Now before clicking DONE, you have to DRAG the selected device into the cell and drop it where you want it to FIX. Then click DONE or press ESC.

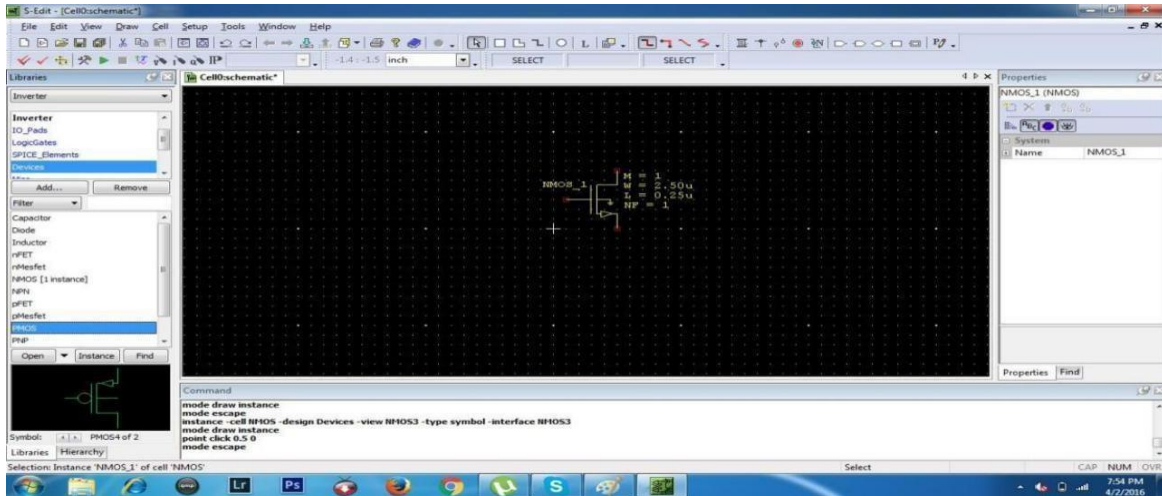


Fig 5.10: Instances of NMOS

STEP 11

Similarly, you can DRAG & DROP any device into the cell for draw your schematic circuit. For inverter we need another PMOS

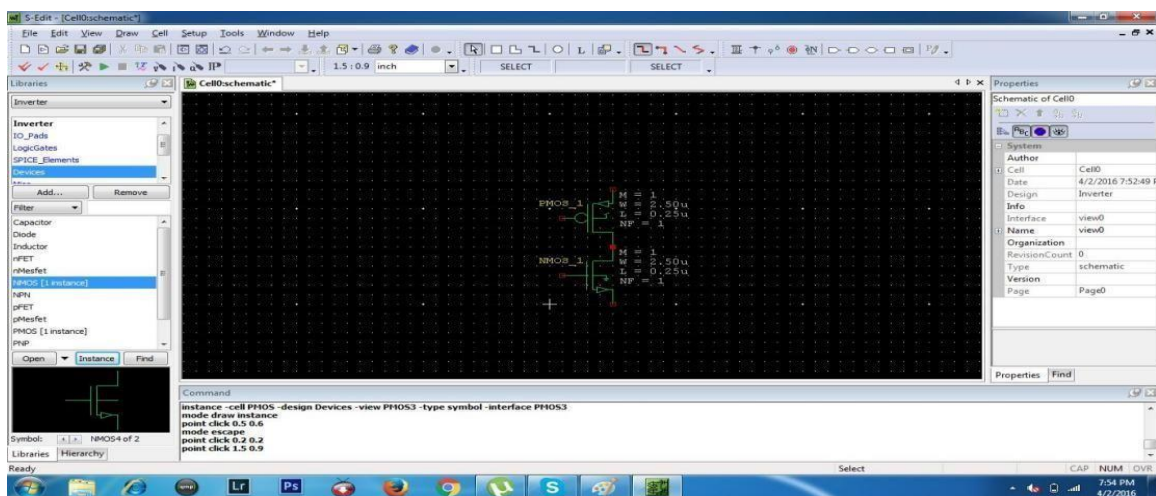


Fig 5.11: Circuit diagram of Inverter

STEP 12

Select the input and output port from the ribbon

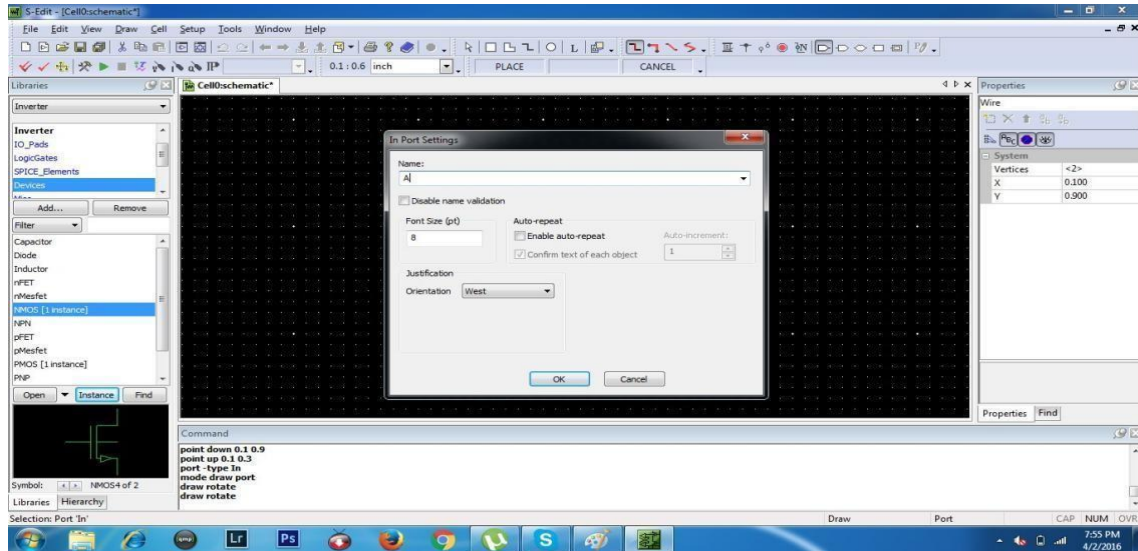


Fig 5.12: Naming the input and output port

STEP 13

Now, after completed these steps, you should give the supply (VDD) & ground.

For that Go to libraries >> MISC >> Select VDD and GND

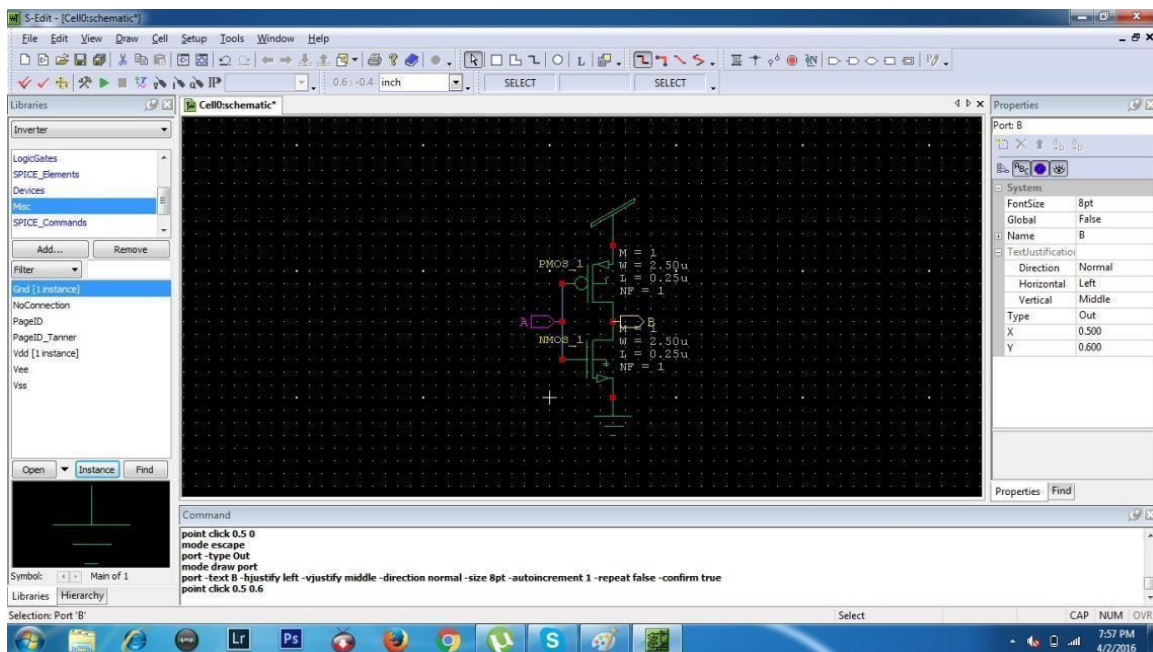


Fig 5.13: GND and VDD Connection

STEP 14

Now check for any errors from the icon shown below

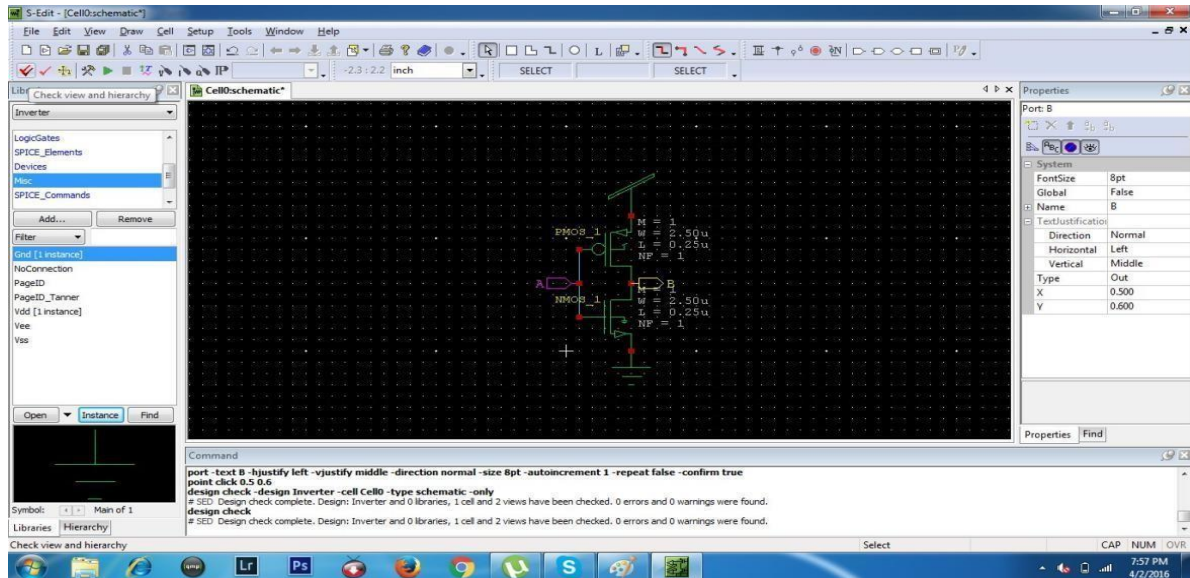


Fig 5.14: Checking for design errors

5.3. SIMULATION

After schematic design you have to check whether your design match with the specification or not. That's why you need to simulate the design which is called simulation.

STEP 15

Generate the netlist by clicking the Tools>>T-spice

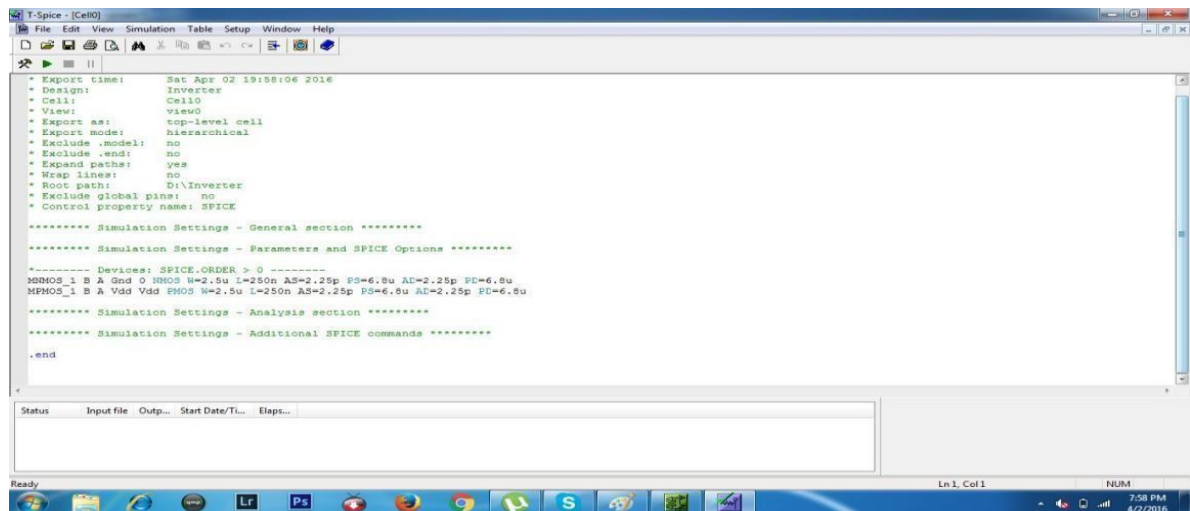


Fig 5.15: Generating netlist

STEP 16

A T-spice window will open. Then click on the bar shown by red ellipse

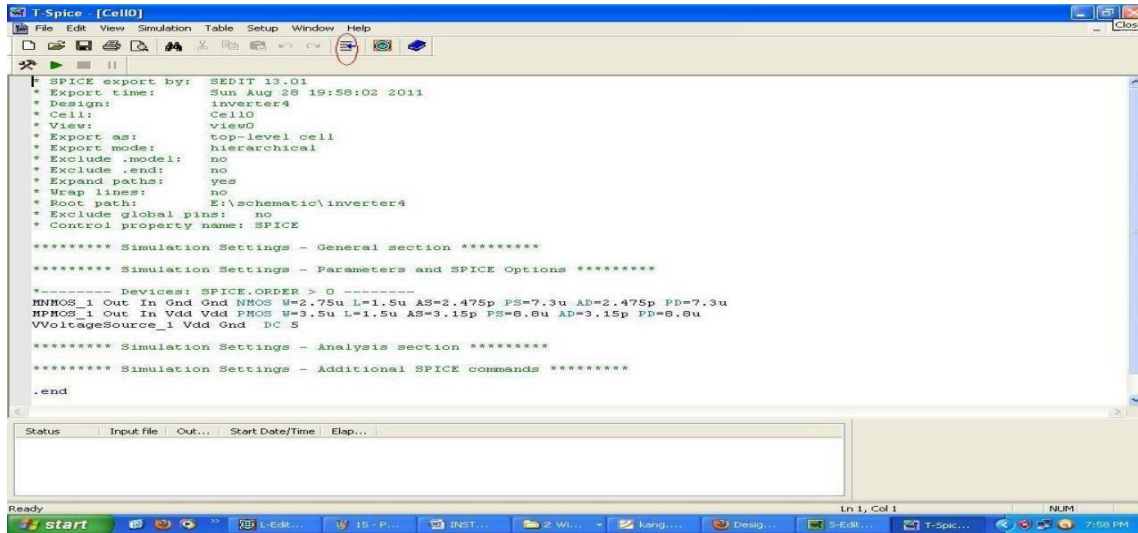


Fig 5.16: For inserting a command

STEP 17

A “T-spice command Tool” dialog box will open as shown below. On the T-spice command you can see in the left-hand side Analysis, Current source Files Initialization, Output Settings Table Voltage source Optimization

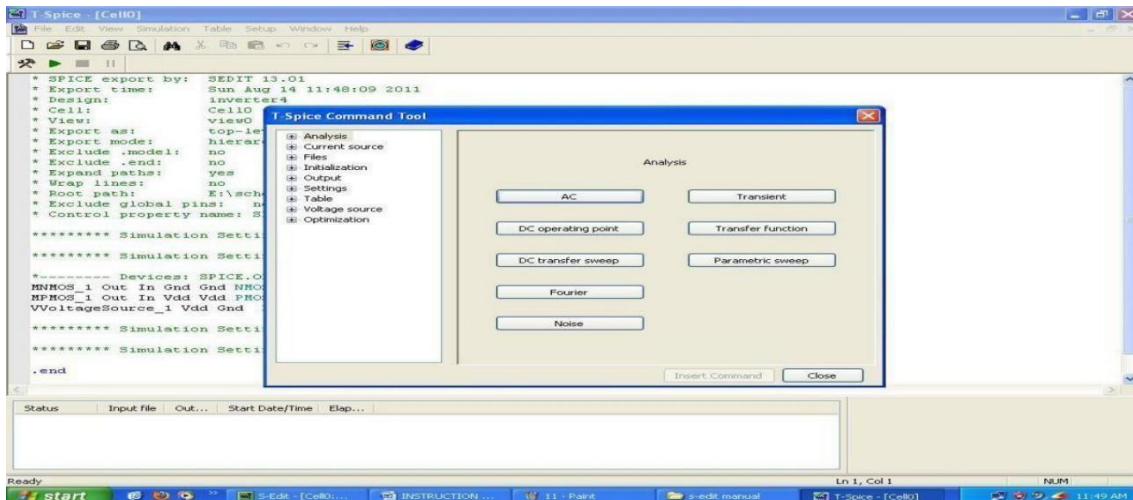


Fig 5.17: Dialog box of Insert Command

STEP 18

Go to voltage source then a dialog box appears having many options.

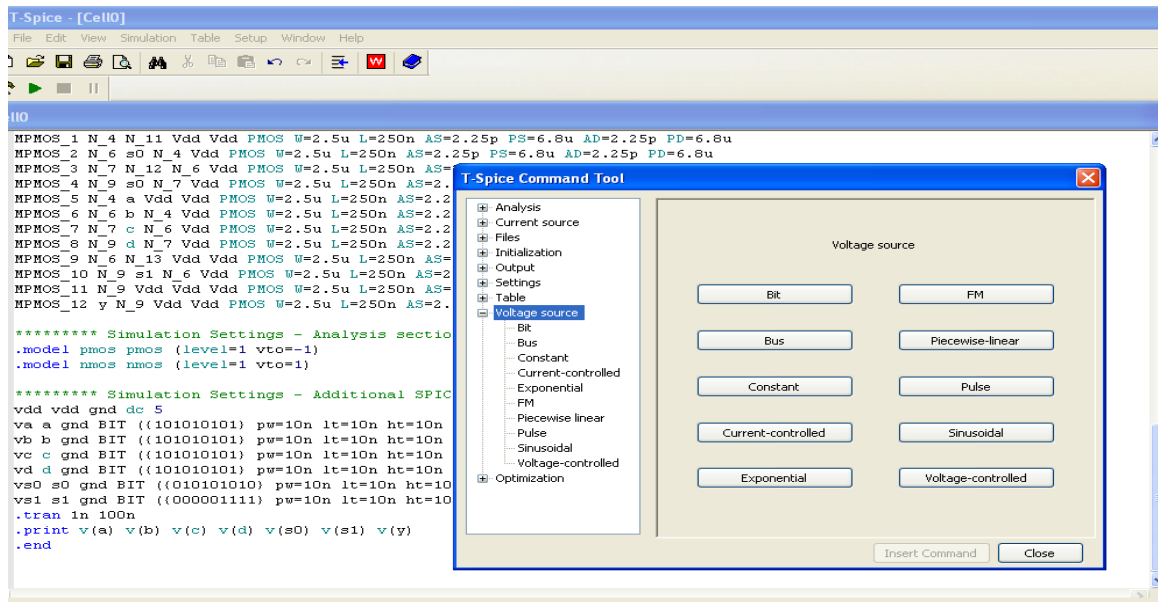


Fig 5.18: Dialog Box of voltage source

STEP 19

click on BIT and enter the input voltage source, input bit and the rise time, fall time, pulse width low time, high time are entered

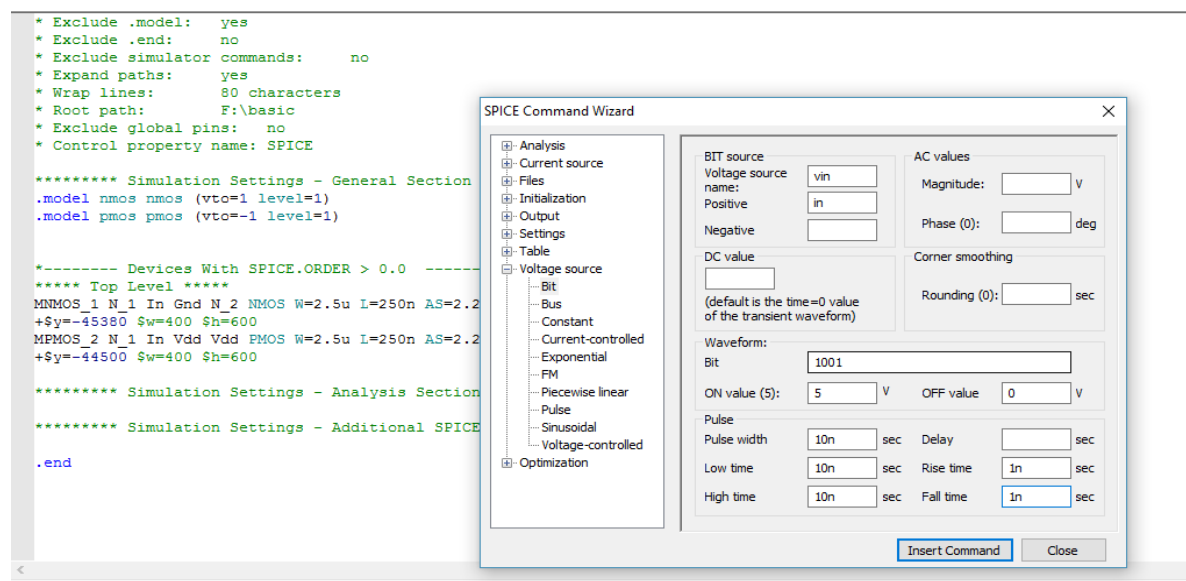
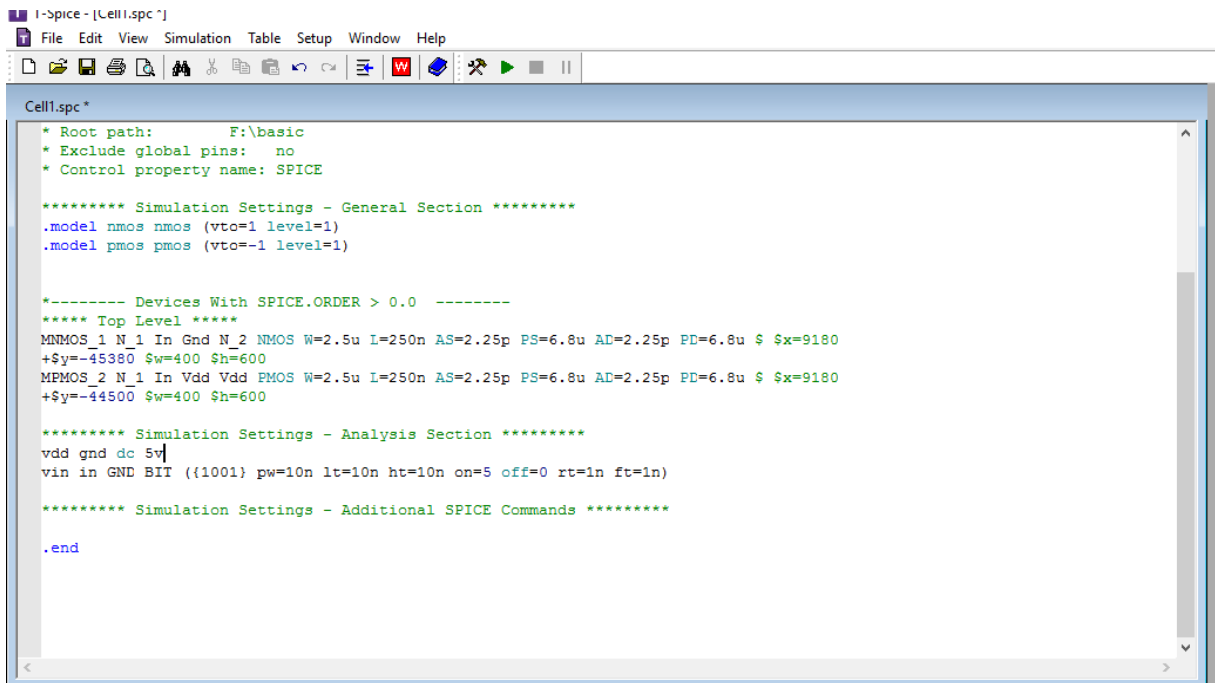


Fig 5.19: Dialog box of BIT

STEP 20

The command is inserted in the T spice netlist



```
I-spice - [Cell1.spic*]
File Edit View Simulation Table Setup Window Help

Cell1.spic*
* Root path:      F:\basic
* Exclude global pins:  no
* Control property name: SPICE

***** Simulation Settings - General Section *****
.model nmos nmos (vto=1 level=1)
.model pmos pmos (vto=-1 level=1)

*----- Devices With SPICE.ORDER > 0.0 -----
***** Top Level *****
NMOS_1 N_1 In Gnd N_2 NMOS W=2.5u L=250n AS=2.25p PS=6.8u AD=2.25p PD=6.8u $ $x=9180
+$y=-45380 $w=400 $h=600
PMOS_2 N_1 In Vdd Vdd PMOS W=2.5u L=250n AS=2.25p PS=6.8u AD=2.25p PD=6.8u $ $x=9180
+$y=-44500 $w=400 $h=600

***** Simulation Settings - Analysis Section *****
vdd gnd dc 5v
vin in GND BIT ({1001} pw=10n lt=10n ht=10n on=5 off=0 rt=1n ft=1n)

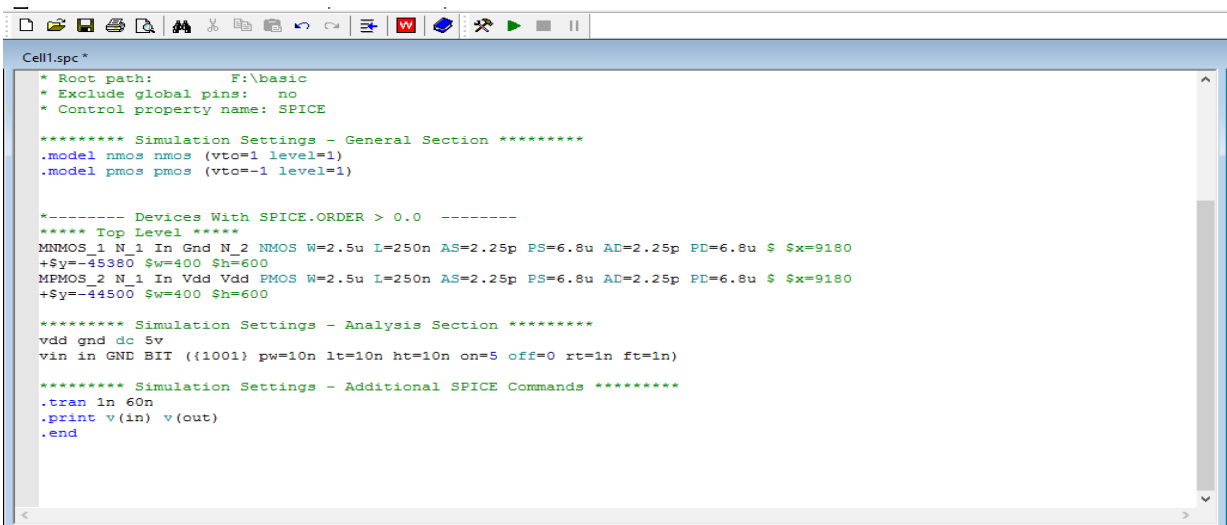
***** Simulation Settings - Additional SPICE Commands *****

.end
```

Fig 5.20: Netlist after inserting command

STEP 21

Then write TRAN 1n 60n for knowing the transient response of the circuit. and PRINT V(in) V(out) to print the waveforms.



```
I-spice - [Cell1.spic*]
File Edit View Simulation Table Setup Window Help

Cell1.spic*
* Root path:      F:\basic
* Exclude global pins:  no
* Control property name: SPICE

***** Simulation Settings - General Section *****
.model nmos nmos (vto=1 level=1)
.model pmos pmos (vto=-1 level=1)

*----- Devices With SPICE.ORDER > 0.0 -----
***** Top Level *****
NMOS_1 N_1 In Gnd N_2 NMOS W=2.5u L=250n AS=2.25p PS=6.8u AD=2.25p PD=6.8u $ $x=9180
+$y=-45380 $w=400 $h=600
PMOS_2 N_1 In Vdd Vdd PMOS W=2.5u L=250n AS=2.25p PS=6.8u AD=2.25p PD=6.8u $ $x=9180
+$y=-44500 $w=400 $h=600

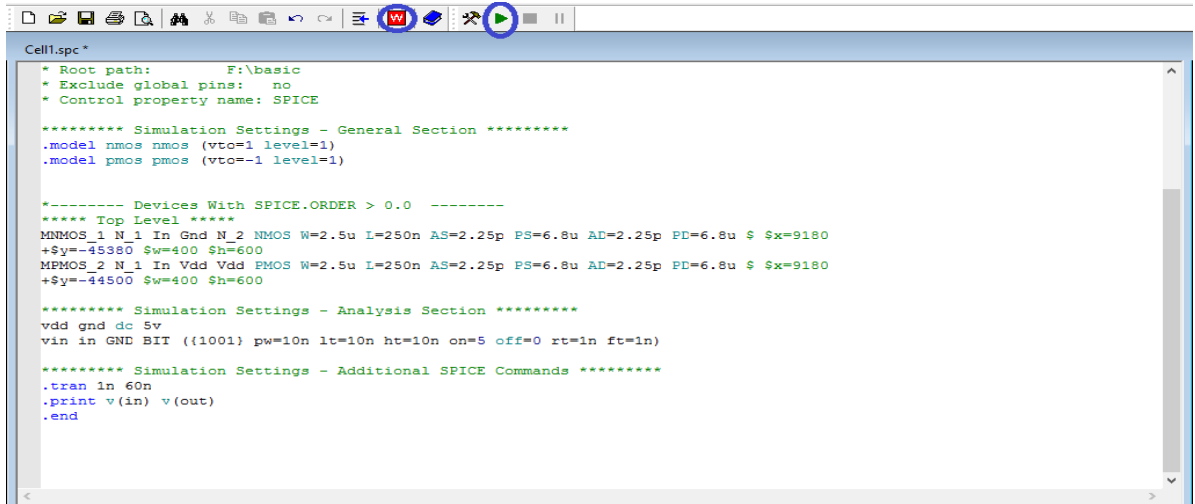
***** Simulation Settings - Analysis Section *****
vdd gnd dc 5v
vin in GND BIT ({1001} pw=10n lt=10n ht=10n on=5 off=0 rt=1n ft=1n)

***** Simulation Settings - Additional SPICE Commands *****
.tran 1n 60n
.print v(in) v(out)
.end
```

Fig 5.21: Netlist for transient response

STEP 22

Click on the RUN option to run the netlist and the "W" option to view the waveforms which are marked by the blue color circles



```
Cell1.spic *
* Root path:      F:\basic
* Exclude global pins:  no
* Control property name: SPICE

***** Simulation Settings - General Section *****
.model nmos nmos (vto=1 level=1)
.model pmos pmos (vto=-1 level=1)

***** Devices With SPICE.ORDER > 0.0 *****
***** Top Level *****
NMOS_1 N_1 In Gnd N_2 NMOS W=2.5u L=250n AS=2.25p PS=6.8u AD=2.25p PD=6.8u $ $x=9180
+$y=-45380 $w=400 $h=600
PMOS_2 N_1 In Vdd Vdd PMOS W=2.5u L=250n AS=2.25p PS=6.8u AD=2.25p PD=6.8u $ $x=9180
+$y=-44500 $w=400 $h=600

***** Simulation Settings - Analysis Section *****
vdd gnd dc 5v
vin in GND BIT ((1001) pw=10n lt=10n ht=10n on=5 off=0 rt=1n ft=1n)

***** Simulation Settings - Additional SPICE Commands *****
.tran in 60n
.print v(in) v(out)
.end
```

Fig 5.22: Final netlist

STEP 23

The window showing the waveforms.

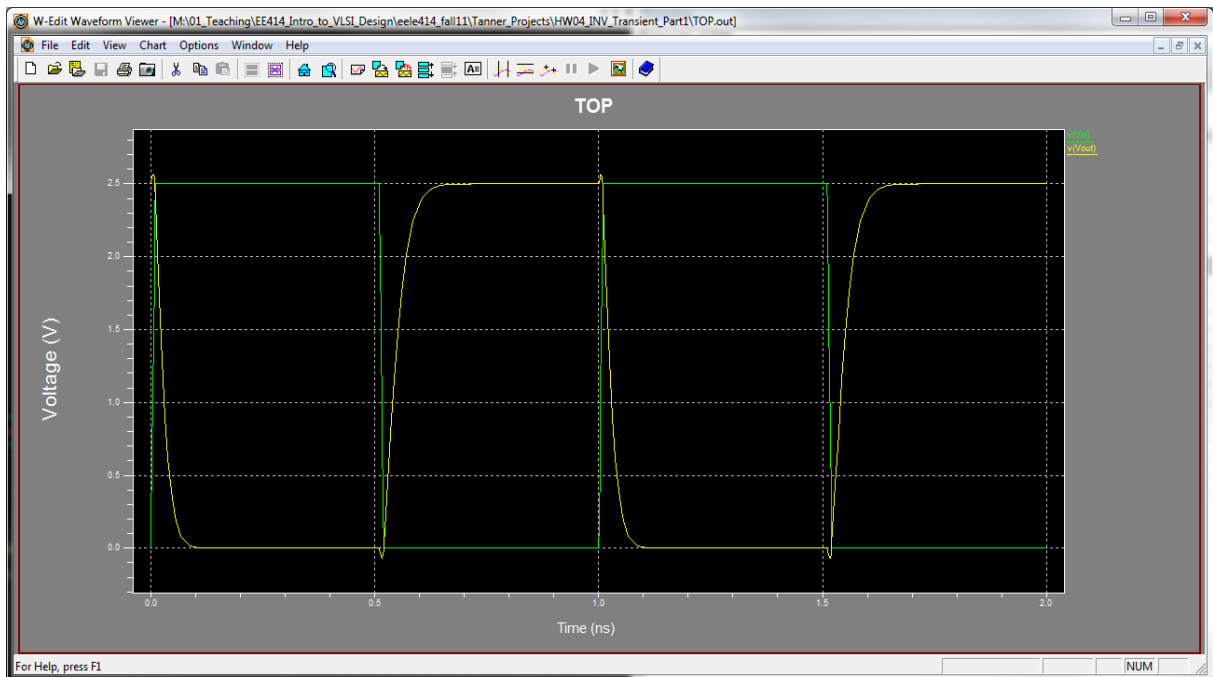


Fig 5.23: waveforms of an inverter

CHAPTER 6

RESULTS

SIMULATION RESULTS

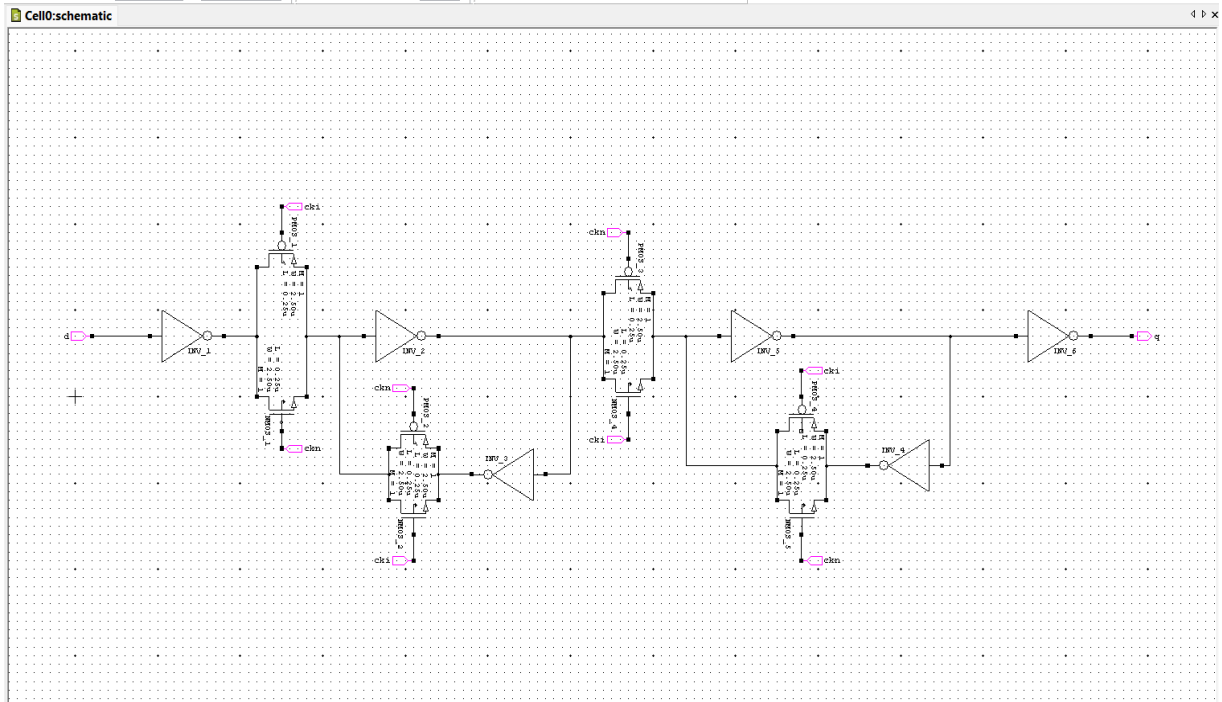


Fig 6.1: Schematic Diagram of Transmission-Gate Flip-Flop (TGFF)

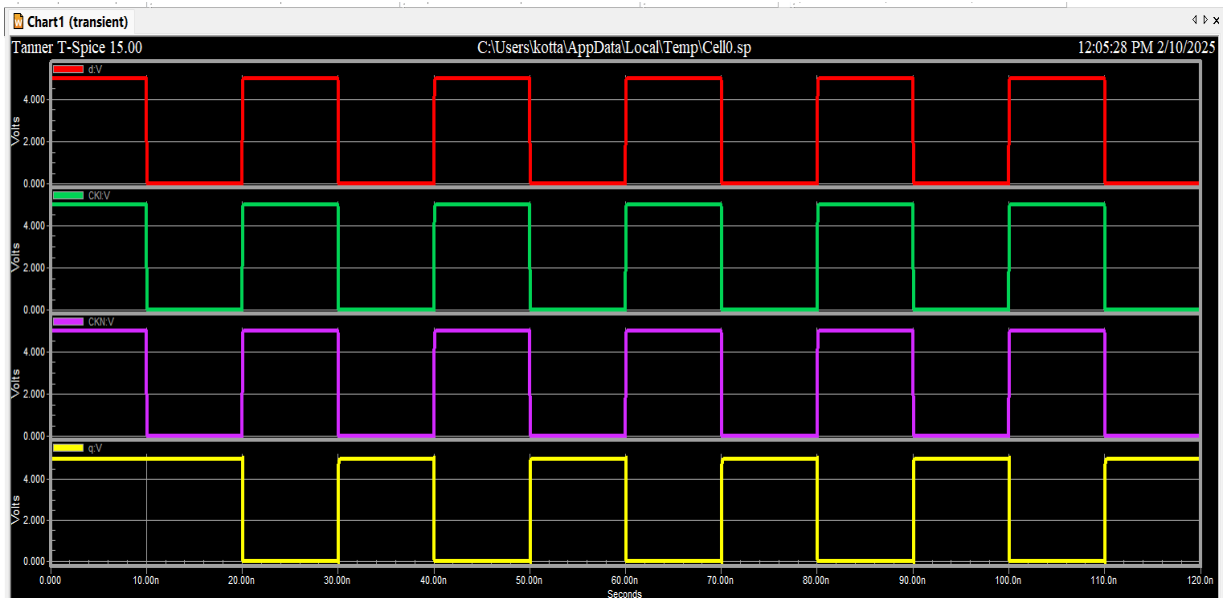


Fig 6.2: Output Waveform of Transmission-Gate Flip-Flop (TGFF)

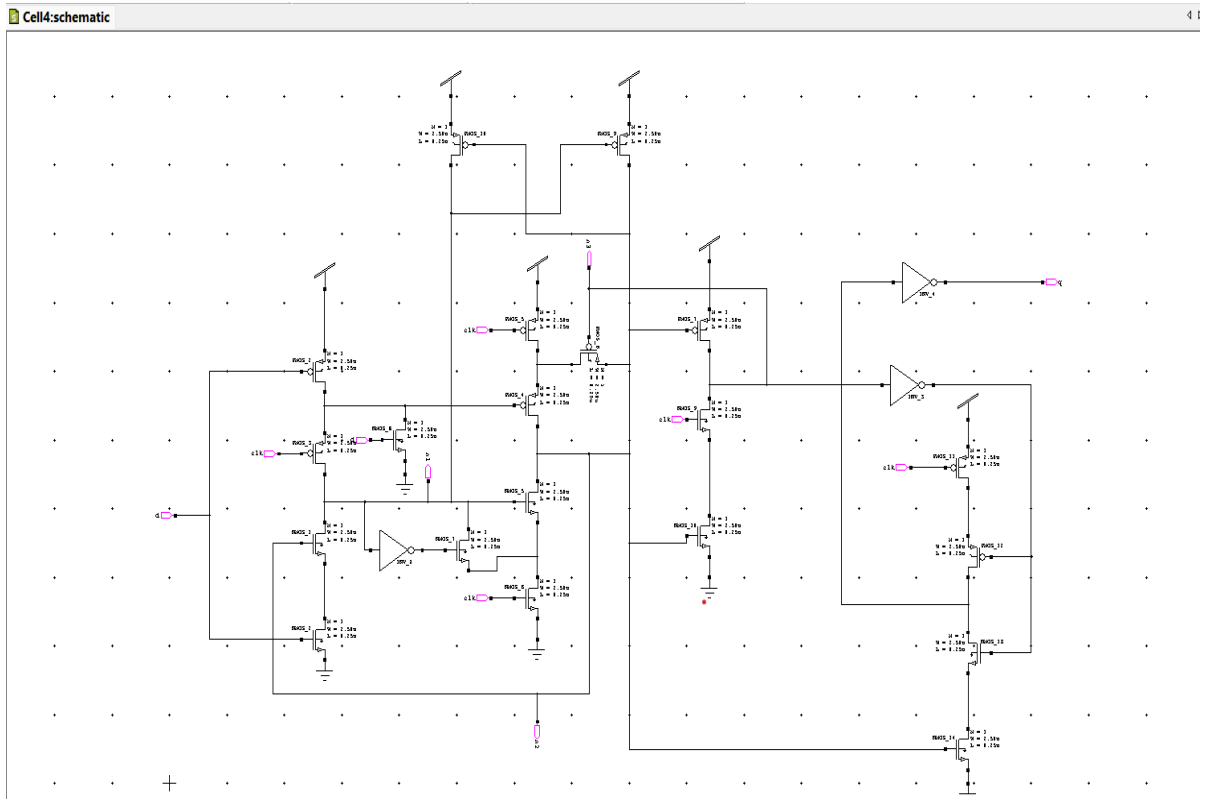


Fig 6.3: Schematic Diagram of True Single Phase Clocked FF (TSPCFF)

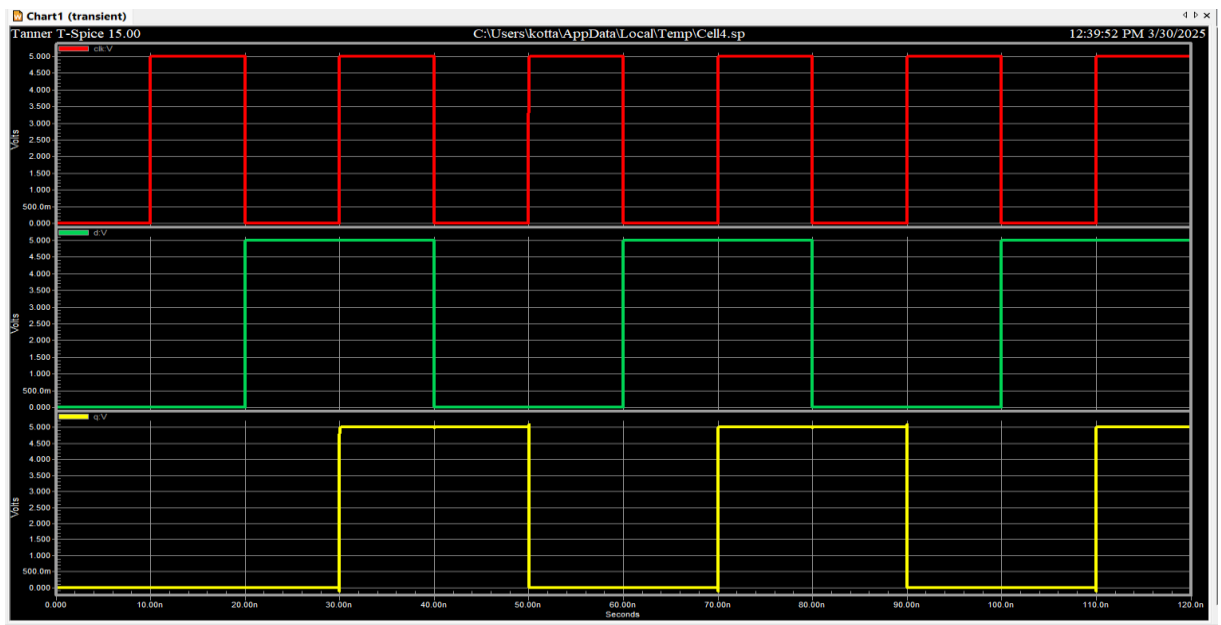


Fig 6.4: Output Waveform of True Single Phase Clocked FF (TSPCFF)

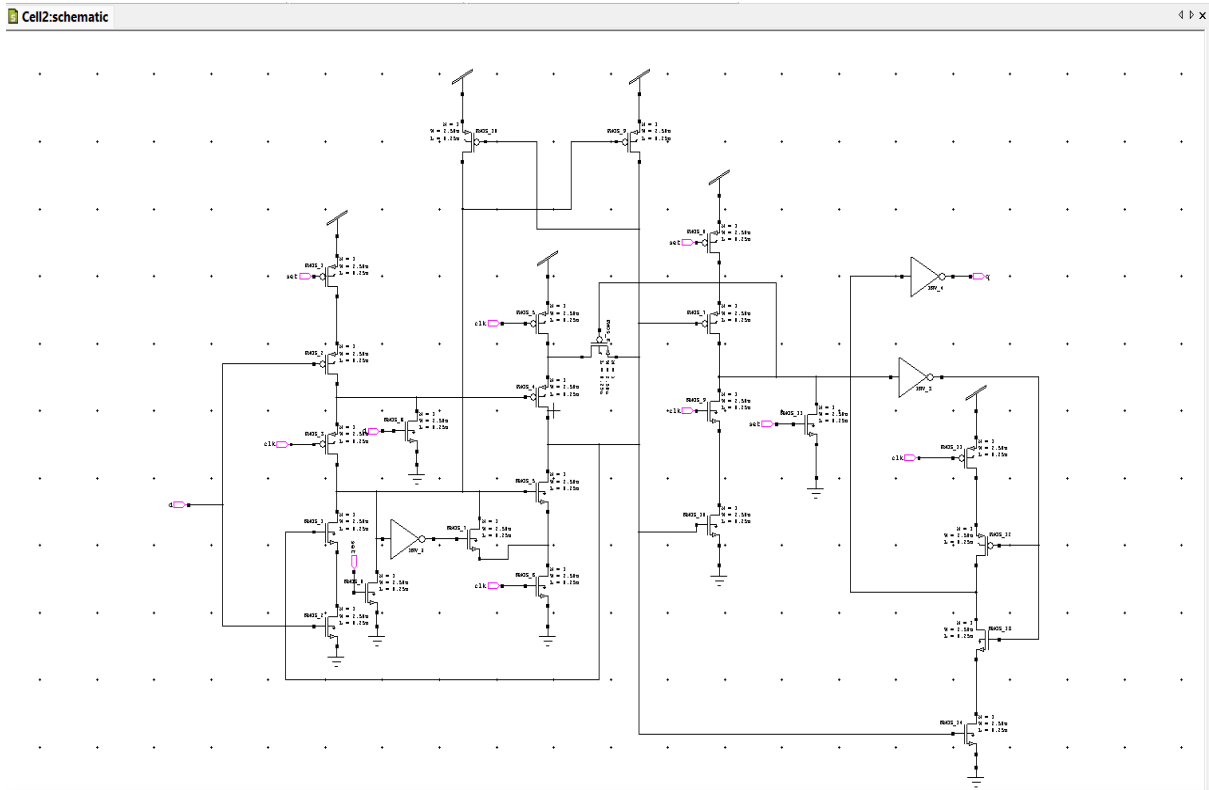


Fig 6.5: Schematic Diagram of True Single Phase Clocked FF (TSPCFF) with Set

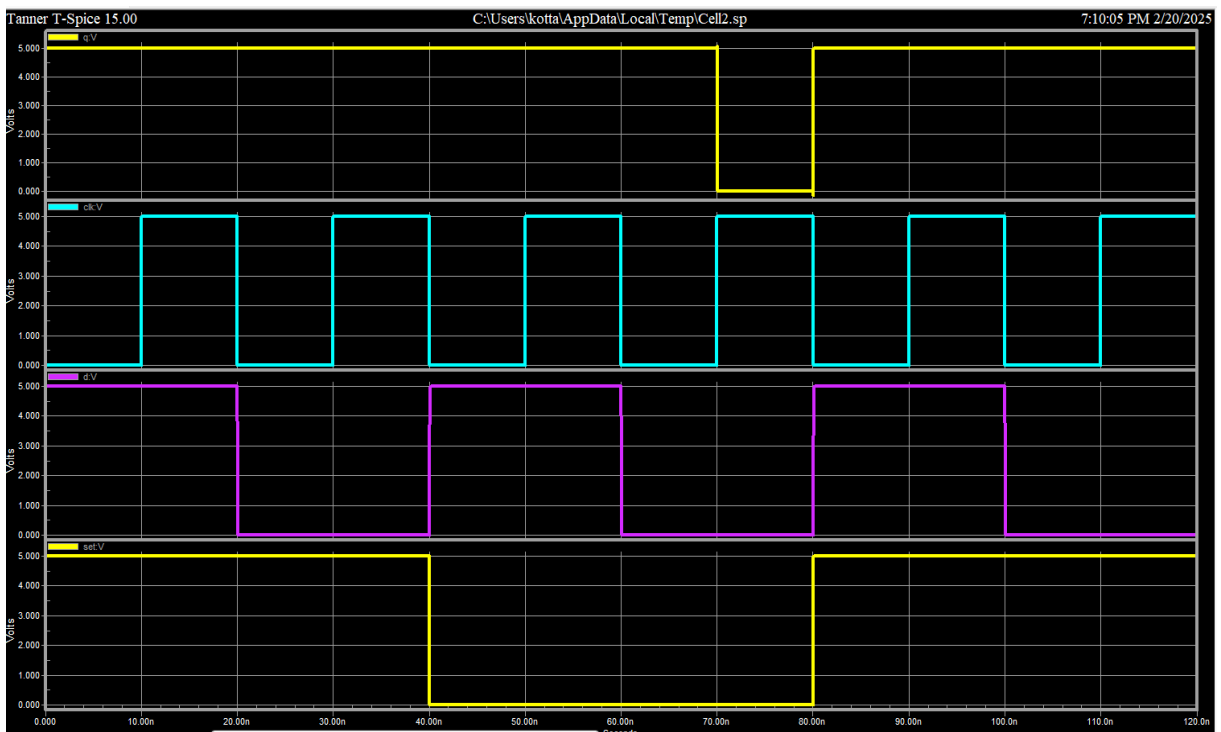


Fig 6.6: Output Waveform of True Single Phase Clocked FF (TSPCFF) with Set

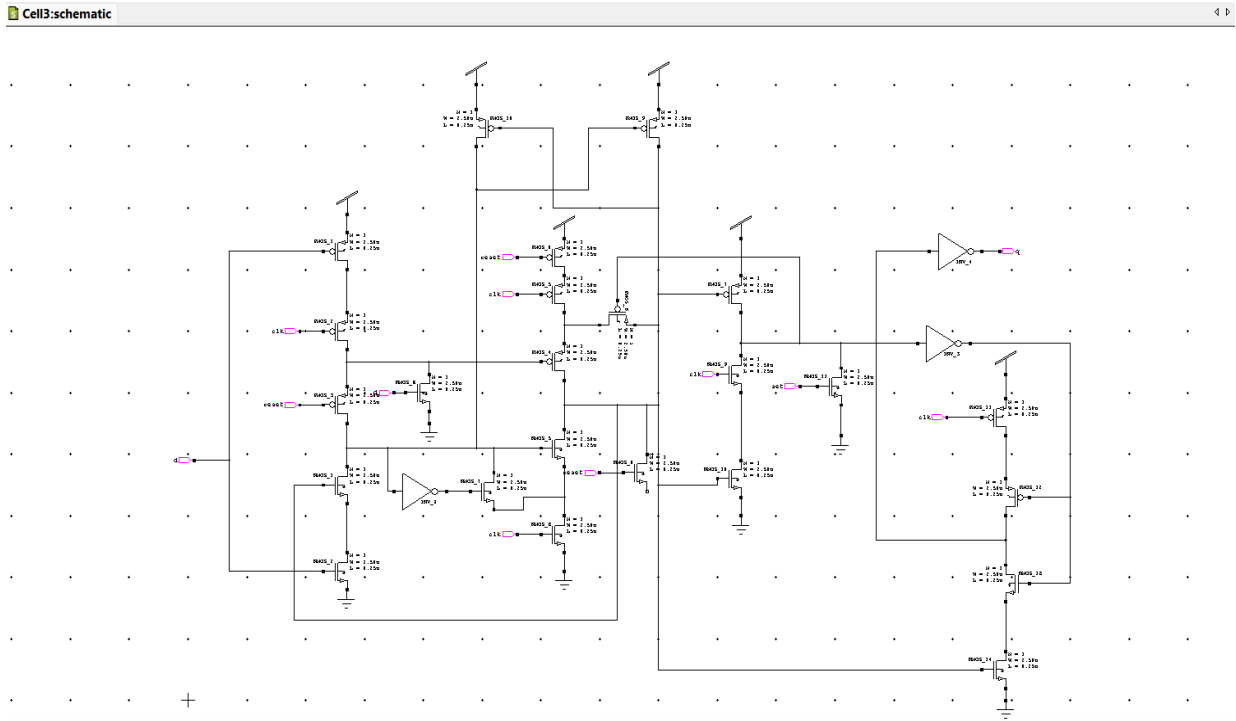


Fig 6.7: Schematic Diagram of True Single Phase Clocked FF (TSPCFF) with Reset

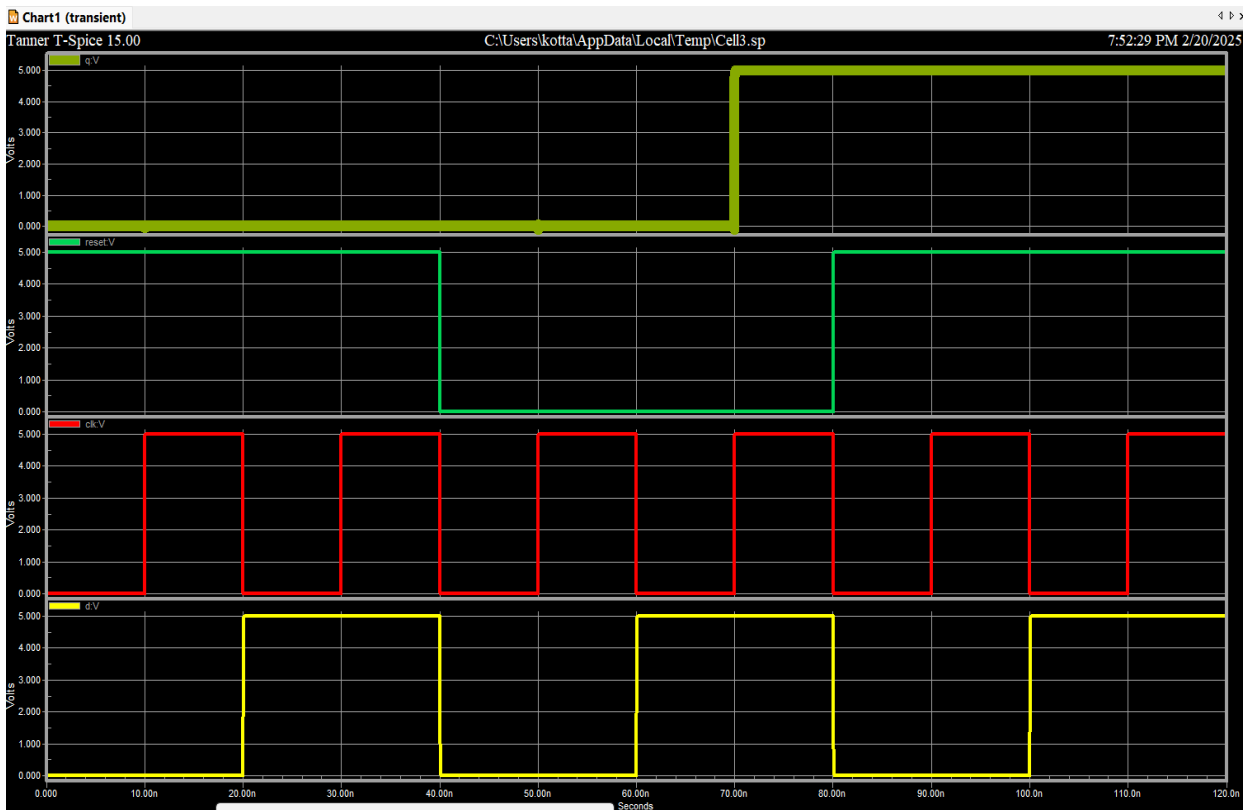


Fig 6.8: Output Waveform of True Single Phase Clocked FF (TSPCFF) with Reset

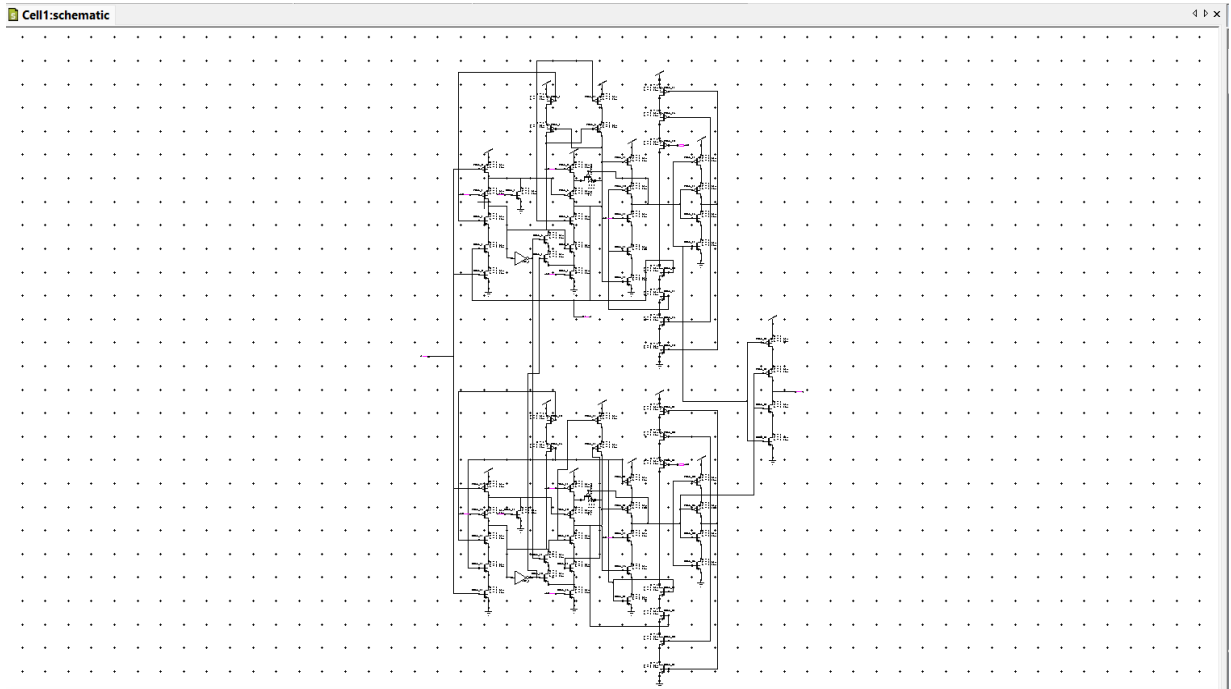


Fig 6.9: Schematic of the Proposed SEU-Tolerant FF (SEU-TFF)

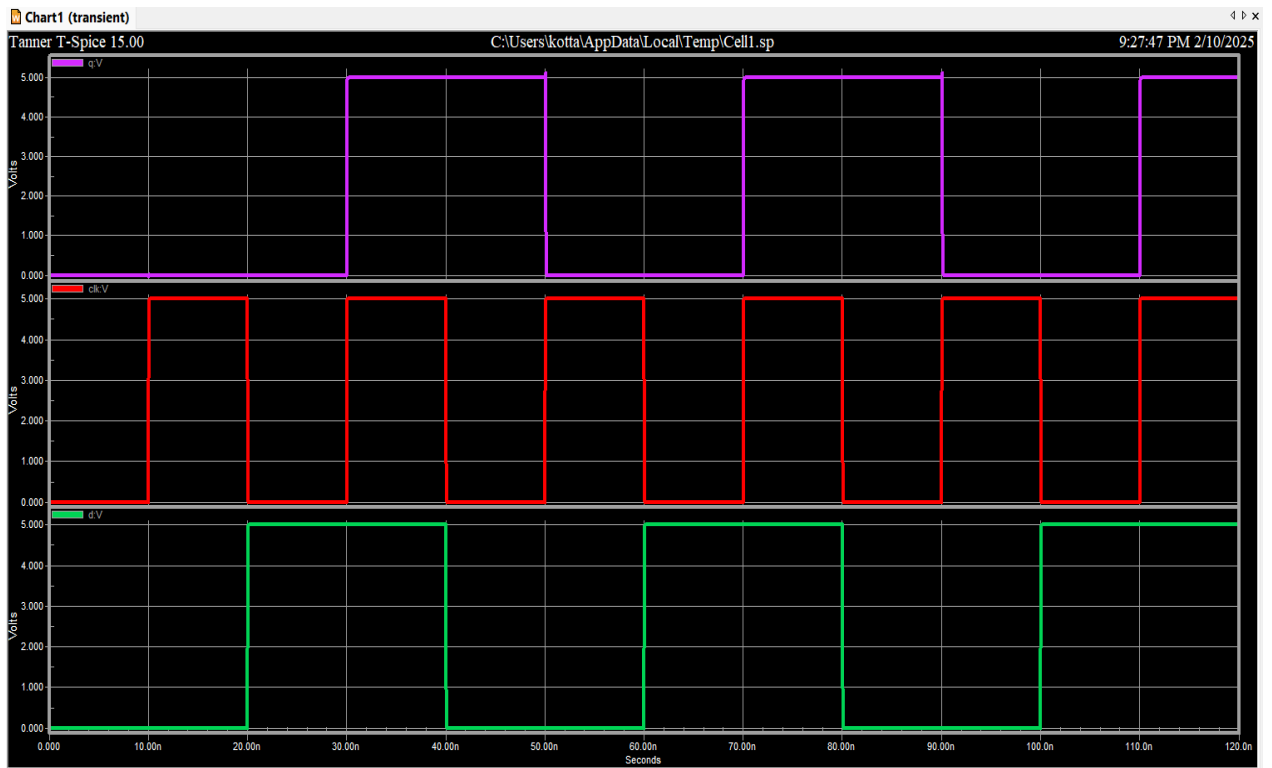


Fig 6.10: Output Waveform of Proposed SEU-Tolerant FF (SEU-TFF)

RESULTS AND DISCUSSION

The proposed ff significantly improves energy efficiency by eliminating redundant pre-charge operation. When data activity attenuates to an absolute minimum, the proposed Flip-Flop (FF) dissipates the power expended by the Transmission Gate Flip-Flop (TGFF), as it requires no extra operations and only consumes leakage power. In contrast, TGFF continues operating its internal nodes (CKN and CKI), leading to unnecessary power consumption. The proposed FF achieves power savings compared to TGFF at various supply voltages.

Additionally, its delay is 26.18% lower at a supply voltage of 1.2V. However, the setup time of the proposed FF is slightly higher due to its data-dependent pre-charge operation, where multiple transistors (M1, M15, M6) are involved in the process. The hold temporal parameter of the postulated Flip-Flop (FF) constitutes approximately negligible, analogous into the Transmission Gate Flip-Flop (TGFF). Within the framework of the Flip-Flop (FF) incorporating a reset function, the augmented capacitive burden on node n2 marginally elevates the clock-to-output (CK-to-Q) delay, while the diminished efficiency of the pre-charge path at N2 extends the initialization interval in the Flip-Flop (FF), CK-to-Q delay remains unaffected; however, the setup time increases due to the attenuation of the pull-down networks at N1 and M1.

Hence the True-Single-Phase-Clocked (TSPC) flip-flop has been designed to operate with lower energy consumption. This optimized TSPC flip-flop incorporates an input-aware pre-charge strategy, which ensures that pre-charging occurs only when necessary, thereby minimizing unnecessary power dissipation. Additionally, transistor-level optimizations and floating node analysis have been employed to further enhance energy efficiency without significantly increasing the circuit's area. These optimizations help reduce leakage power and unnecessary switching activity, leading to improved performance.

Parameters	TSPC	TSPC with set	TSPC with reset	SEU
Power	2.02 μ W	4.5 μ W	4.5 μ W	2.9 μ W
Delay	17.5091sec	40.0055sec	40.0055sec	17.5223sec

Table 6.11 : Power and Delay Calculations

The analysis of performance for various TSPC architectures provides useful information about the impact of incorporating additional functionalities on the key design parameters such as power consumption and delay. The minimum TSPC flip-flop is the most power-efficient with a power consumption of only 2.02 μW , and a delay of 17.5091 ns, which is best suited for high-speed and low-power applications. But the moment a set functionality is introduced into the design, the power and the delay register a significant increase. The TSPC with Set version of the design has a power consumption of 4.5 μW and a delay of 40.0055 ns, which indicates that this additional functionality comes at the cost of performance.

Likewise, the TSPC with Reset architecture also consumes the same power of 4.5 μW but with significantly higher delay of 69.9938 ns. This indicates that the reset mechanism adds more complexity, which leads to a dramatic decrease in speed. Nevertheless, the SEU-tolerant flip-flop design offers an optimal solution with moderate power consumption of 2.9 μW and delay of 17.5223 ns. This renders it suitable for systems that demand resistance to soft errors without compromising on acceptable performance levels. Overall, this comparison helps choose the most suitable flip-flop design according to the particular demands for power efficiency, speed, and reliability.

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

CONCLUSION

An energy-efficient retentive TSPC FF is suggested in this paper. The input-aware pre-charge approach significantly lowers the power of the suggested FF by eliminating unnecessary pre-charge and discharge operations. Additionally, the suggested topology uses examination of floating nodes to stop the development of short-circuit paths. To further reduce the power and area consumption, the circuit is further tuned at the transistor level. Based on the outcomes of the post layout simulation, under 10% data activity, the suggested FF uses about 80% less power than TGFF. Ten test chip measurements also show that the suggested FF significantly improves energy efficiency when compared to TGFF. The suggested FF's CK-to-Q latency is 26.18% less than TGFF's. There is minimal area overhead required to realize these gains, as the proposed FF's area is only 4.8% greater than TGFF's.

FUTURE SCOPE

Hence, TSPC logic d flip-flop is less power consuming than other flip-flops. The circuit is then optimized at the transistor level to further reduce its size and power usage. According to post layout simulation results, the suggested FF consumes less power than TGFF

- Further Power Optimization – Exploring alternative circuit techniques to further minimize power consumption in flip-flop designs.
- Technology Scaling – Analyzing how the proposed design performs in smaller technology nodes (e.g., 5nm, 3nm).
- Application in IoT and AI Chips – Integrating the design in real-world applications like IoT devices or low-power AI accelerators.

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CO-PO Mapping Table

Project Title : Energy-Efficient True Single Phase Clocked Flip Flop for High Performance Applications.

Batch No : B13

Student Names: 1. K. Sailatha (21NM1A0475)

2. K. Bhoomika (21NM1A0479)

3. K. Niharika (21NM1A0471)

4. M. Rupavathi (21NM1A0484)

Project Course Outcomes:

CO1: Analyze the limitations of existing flip-flop architectures and identify key design challenges in energy efficiency.

CO2: Design and develop a low-power TSPC flip-flop using optimized transistor sizing and clocking strategies.

CO3: Implement and simulate the flip-flop circuit using Tanner EDA tools to verify functionality and performance.

CO4: Evaluate power consumption, delay, and reliability through theoretical, practical analysis.

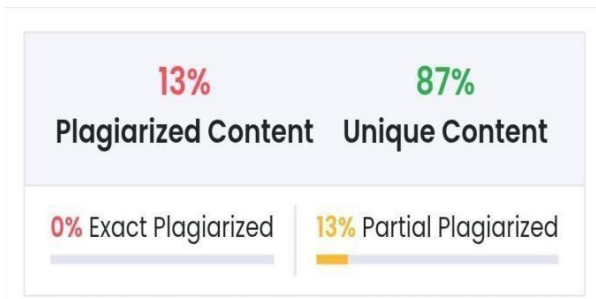
CO5: Compare the proposed flip-flop design with conventional architectures and explore potential applications in high-speed circuits.

CO-PO Mapping:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	3	3	2	2	-	-	-	-	-	-	-	-	3	2
CO2	3	3	3	2	2	-	-	-	-	-	-	-	3	3
CO3	3	2	3	3	2	-	-	-	-	-	-	-	3	3
CO4	3	2	2	3	3	-	-	-	-	-	-	-	3	3
CO5	3	3	2	3	2	-	-	-	-	-	-	-	3	3

Plagiarism Report

Chapter 1



Chapter 2



Chapter 3



Chapter 4



Chapter 5



Total Average Plagiarism:7.5%

Paper Status

